

Kidney & Upper-Tract Tumours

Rapid Revision

Campbell-Walsh-Wein · Chapters 136-139, 141-144

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 4½-5 h at a slow, careful pace)

- **Sections 1-5 of each chapter:** read for understanding. Tables are your answer diagrams.
- **Theory answer frames:** headings in the order to write them; flesh out from the tables.
- **Viva corner:** cover the answer, say yours aloud, then check.
- **MCQ one-liners:** rapid re-read the day before the paper.
- Last section: cross-chapter tables + one-page night-before sheet.

Key: **number to memorise** · **MCQ** exam-trivia only · → [Ch 141](#) cross-reference ·
outside the source notes

Chapter	Topic	Slow-reader time
136	Benign Renal Tumours	19 min
137	Malignant Renal Tumours	52 min
138	Urothelial Tumours of the Upper Tract & Ureter	27 min
139	Surgical Management of UTUC	33 min
141	Contemporary Open Surgery of the Kidney	33 min
142	Laparoscopic & Robotic Surgery of the Kidney	29 min
143	Focal Therapy for Renal Tumours	20 min
144	Treatment of Advanced RCC	34 min
★	Cross-chapter tables & night-before sheet	12 min
Total (18080 words)		4h 19min

136 Benign Renal Tumours

Reading time ≈ 19 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★★

1 · Core concepts

- Up to **~25%** of resected SRMs benign. No modality (US/CT/MRI) reliably separates benign from malignant.
- Imaging-diagnosable benign lesions: **simple cyst** and **AML** (macroscopic fat).
- Management scales with RCC risk: none (simple cyst) → SAE (large AML) → excision (solid/complex cystic). Biopsy & SRM algorithm → [Ch 137](#).

2 · Renal cysts & Bosniak

- Commonest renal mass; ↑ age, male, renal dysfunction; grow 1–2 mm/yr.
- Simple cyst**: homogeneous, **≤20 HU** → no further imaging. Heterogeneity, >20 HU, septa, thick wall, nodule, calcification → Bosniak.

Class	Features (v2019, CT/MRI)	Malignancy	Management (CUA 2023)
I	Thin (≤2 mm) smooth wall (may enhance); simple fluid; no septa/Ca	Benign	No follow-up
II	Thin wall + (a) ≤3 thin enhancing septa, any calcification; or (b) homogeneous marked T2 ↑; or (c) marked T1 ↑	Highly likely benign	No follow-up
IIF	Minimally thickened (3 mm) enhancing wall/septum, or ≥4 thin septa; or heterogeneous T1 ↑ on fat-sat MRI	Likely benign	Surveillance up to 5 yr
III	Enhancing thick (≥4 mm) or irregular (≤3 mm obtuse protrusion) wall/septum	~50%	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, PN > RN
IV	Enhancing nodule: ≥4 mm obtuse or any-size acute-margined protrusion	~90%	

Ladder: wall ≤2 → 3 → ≥4 mm → irregularity → nodule. Traps: any calcification OK in II; I wall may enhance; ≥3 cm hyperdense cyst no longer IIF in v2019. EAU: IV = manage as solid mass; biopsy if solid component and result alters management. Ablation for cysts: uncertain.

- v2019 added MRI, mainly because III was over-treated (low PPV; very high NNT to avoid metastasis).
- Cystic RCC indolent (volume largely fluid) → proposed **pT1c**. Size, not category alone, drives III–IV choice.

	ADPKD	ACKD
Cause	PKD1 (chr 16, polycystin-1), PKD2 (chr 4)	ESRD/dialysis; PCT origin; 80% by 10 yr
Kidneys	Huge, distorted; cause of ESRD	Small/normal; consequence of ESRD
RCC	Not ↑ risk; earlier, more sarcomatoid, harder to diagnose	≥10× ; multifocal, bilateral, male, papillary; WHO 2022 entity

- ADPKD: extrarenal cysts (liver, pancreas, spleen, ovary, SV, arachnoid). US diagnosis: PKD1 relative 15–29 yr, **≥3 cysts**; exclusion secure only ≥40 yr (<2 cysts). **Tolvaptan** (V2 antagonist) slows decline.
- Urology: stones, pain, haematuria/cyst bleed (self-limiting), cyst infection (FDG-PET; fluoroquinolone/TMP-SMX). Aspiration, embolisation, fenestration, nephrectomy only selectively; ESRD → transplant.

3 · Oncocytoma

- Commonest benign enhancing mass**; up to half of benign SRMs <4 cm.
- Syndromic: **Birt-Hogg-Dubé** (AD, folliculin, chr 17): fibrofolliculomas, lung cysts/pneumothorax, oncocytoma + chRCC. Oncocytosis: diffuse bilateral nodules → CKD; may harbour chRCC/hybrid tumours.
- Imaging: central **stellate scar** (non-specific); **segmental enhancement inversion** favours it. MRI T1 ↓ T2 ↑. Sestamibi SPECT: not yet routine.

- Core biopsy cannot definitively diagnose (EAU): sampling error + oncocyoma-like areas in chRCC.

	Oncocyoma	Chromophobe RCC
CK7	Rare, single-cell; KIT+	Diffuse +
Cytogenetics	Normal, -1, -Y, 11q13 rearrangement	Extensive

4 · Angiomyolipoma

- PEComa: dysmorphic vessels + smooth muscle + fat. Female, 4th-5th decade. In TSC: **55-90%**.
- TSC2** (tuberin, 16p13) > TSC1 (hamartin, 9q34) biallelic loss → mTOR disinhibition. Multiple tumours clonally independent.
- IHC: **HMB-45, Melan-A**, SMA +; keratin -.

	Sporadic	TSC/LAM	Epithelioid
Lesions	Solitary, small	Bilateral, multifocal, recurrent; bleed, HTN, CKD	Fat-poor; mitoses, necrosis, venous extension, mets
Follow-up	US/CT 6, 12 mo, then yearly	MRI 1-3 yr	—
Treat	Triggers (below)	Everolimus if >3 cm	As RCC

- Imaging: fat **-15 to -20 HU** on NCCT diagnostic; US unreliable (hyperechoic RCC); MRI India-ink artefact, fat-sat drop-out. Fat-poor AML = pitfall.
- Giant AML vs liposarcoma: AML arises from parenchyma; liposarcoma envelops/compresses kidney.

No fat → MRI/biopsy (RCC differential)

Spontaneous retroperitoneal haemorrhage (Wunderlich) → resuscitate → **SAE** → RN if fails/SAE unavailable

Symptomatic (pain, haematuria, barriers to care) → SAE → NSS/RN → ablation (selected)

Asymptomatic sporadic: **>6 cm**, aneurysm **>5 mm**, growth **>0.25 cm/yr**, childbearing age/pregnancy → treat; else surveil

- Old **4 cm** trigger: sensitive, poorly specific → overtreatment. <4 cm → surveillance; ≥4 cm observed → thin-slice CTA for aneurysm.
- SAE: first line (acute, symptomatic, prophylactic, pre-op); post-embolisation syndrome common, self-limiting. NSS preferred surgery (crucial in TSC).
- TSC consensus 2021: haemorrhage → embolise + steroids, avoid nephrectomy; growing >3 cm → mTOR inhibitor first line. EAU sporadic: mTOR only if not SAE/surgery candidate.

5 · Other benign tumours

- Papillary adenoma**: ≤ **15 mm** (WHO 2016/22; was ≤5 mm); +7, +17, -Y, KMT2C like papRCC; no metastatic capacity; donor kidney OK.
- Metanephric adenoma**: women, 50s; polycythaemia; hypovascular, mild enhancement; **WT1, CD57** +; **CK7, AMACR** -; lacks +7/+17/-Y; excise by NSS.
- MEST/adult cystic nephroma**: one family (WHO 2022); middle-aged women; ER/PR+ stroma; mimic cystic RCC → Bosniak pathway. Paediatric cystic nephroma: **DICER1**.
- Leiomyoma**: renal capsule; T1/T2 hypointense; SMA, desmin, caldesmon+; **Melan-A negative** (vs lipid-poor AML, both HMB-45±).

Tumour	Key trait
Haemangioma	Sturge-Weber, Klippel-Trénaunay
Haemangioblastoma	VHL; mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
JG cell tumour (reninoma)	Young women; refractory HTN, hypokalaemia, ↑ renin; ≤3 cm; renin granules pathognomonic

Tumour	Key trait
Renomedullary interstitial	Incidental, <2 cm
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trials

Trial	Design	Result	Take-home
EXIST-2	Phase III RCT, everolimus vs placebo, TSC/LAM AML	Response (≥50% volume ↓) 42% vs 0% ; no bleeds/nephrectomy	FDA/EMA approval; mTOR first line in TSC

7 · Theory answer frames

Q: Bosniak classification v2019 and management of complex cysts

1. **Simple cyst** – ≤20 HU; no follow-up.
2. **Table** – classes, thickness ladder, risk.
3. **Management** – CUA size-based; shared decision; pT1c concept.

Q: Management of renal AML

1. **Pathology/genetics** – PEComa, TSC2/TSC1, mTOR.
2. **Imaging** – fat, fat-poor pitfalls.
3. **Algorithm** – haemorrhage, symptoms, triggers (>6 cm, aneurysm, growth, pregnancy).
4. **Options** – SAE, NSS, ablation, everolimus (EXIST-2).

Q: Oncocytoma vs chromophobe RCC

1. **Imaging** – scar, inversion.
2. **BHD**.
3. **Biopsy limits, CK7, cytogenetics**.

Q: Short note — MEST / adult cystic nephroma

1. **Definition** – WHO 2022 family; paediatric CN (DICER1) distinct.
2. **Patient/imaging** – middle-aged women; multiseptated, mimics cystic RCC.
3. **Histology** – ER/PR+ spindle stroma; hobnail lining (ACN).
4. **Management** – Bosniak pathway; consider biopsy; NSS; rare malignant change.

8 · Viva corner

Q. Why was Bosniak updated in 2019?

→ Add MRI; too many benign Bosniak III resections.

Q. Irregularity vs nodule?

→ Obtuse ≤3 mm = III; ≥4 mm obtuse or any acute = IV.

Q. Why surveil small Bosniak III?

→ Low PPV, cystic RCC indolent, very high NNT.

Q. Can biopsy diagnose oncocytoma?

→ Not definitively — chRCC contains oncocytoma-like areas.

Q. Why not 4 cm for AML?

→ Poor specificity; >6 cm proposed; add aneurysm >5 mm, growth.

Q. First line for bleeding AML?

→ SAE; nephrectomy if unstable/SAE unavailable.

Q. Why everolimus in TSC?

→ Targets causative mTOR; lesions multifocal/recurrent → nephron-sparing.

Q. ADPKD vs ACKD RCC risk?

→ ADPKD not raised; ACKD $\geq 10\times$.

Q. Stain separating leiomyoma from lipid-poor AML?

→ Melan-A (+ in AML).

Q. Pregnancy-planning woman with AML?

→ Consider prophylactic treatment — higher bleed risk, graver consequences.

Q. Why is metanephric adenoma often excised?

→ Imaging non-specific; biopsy accuracy unproven → NSS.

Q. Young woman, resistant HTN, hypokalaemia, small renal mass?

→ Reninoma; excise.

9 · MCQ one-liners

1. Simple cyst ≤ 20 HU.
2. Bosniak III ~50%, IV ~90% malignant.
3. ≥ 4 septa = IIF; ≤ 3 = II.
4. IIF surveillance up to 5 years.
5. Proposed cystic RCC stage: pT1c.
6. PKD1 chr 16; PKD2 chr 4.
7. ACKD cysts from proximal tubule.
8. BHD: folliculin, chr 17.
9. Oncocytoma: KIT+, CK7 single-cell.
10. Segmental enhancement inversion → oncocytoma.
11. AML: HMB-45, Melan-A +, keratin –.
12. TSC2 = tuberlin 16p13; TSC1 = hamartin 9q34.
13. AML fat –15 to –20 HU on NCCT.
14. AML aneurysm ≥ 5 mm predicts bleed.
15. Epithelioid AML: treat as RCC.
16. Papillary adenoma ≤ 15 mm.
17. Metanephric adenoma: WT1, CD57 +; polycythaemia.
18. DICER1: paediatric cystic nephroma.
19. Leiomyoma: commonest from renal capsule.
20. Haemangioblastoma: VHL.
21. Commonest benign enhancing SRM: oncocytoma.
22. Tolvaptan: V2-receptor antagonist for rapidly progressive ADPKD.
23. ADPKD cyst infection: fluoroquinolone or TMP-SMX; FDG-PET confirms.
24. Everolimus toxicity: stomatitis, acne, proteinuria, hypercholesterolaemia.
25. Papillary adenoma in donor kidney: not a contraindication.
26. Solitary fibrous tumour malignant if ≥ 15 cm, age ≥ 55 .

137 Malignant Renal Tumours

Reading time ≈ 52 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★★

1 · Core concepts

- RCC = 2–3% of adult cancers; most lethal common urological cancer; M:F **2:1**; peak 55–75 y; **4–6% familial**.
- Incidence ↑ ~3%/y from incidental imaging; ~70% organ-confined at diagnosis; >60% detected incidentally ("radiologist's tumour").
- Risk factors: **tobacco** (most accepted, RR ~1.4, dose-dependent, falls after quitting), **obesity** ("obesity paradox": lower stage), **hypertension** (disease, not drugs); family history 2–4x; moderate alcohol may protect.
- Cell of origin: clear cell & papillary → **proximal tubule**; chromophobe & collecting duct → distal nephron.
- Strongest predictors of malignancy: **male sex (~3x)** and **size**. Benign: 20–30% of ≤4 cm, ~40% of ≤2 cm. Nomograms predict malignancy poorly (C-index 0.55–0.65).
- cT1 mass biology: ~20% benign · ~60% indolent RCC · ~20% potentially aggressive.
- Paediatric: RCC 1–6% of childhood renal tumours (Wilms ~90%); equals Wilms in 2nd decade; commonest = **Xp11/TFE3 translocation RCC**; up to 15% post-chemotherapy. Formal LND at RN recommended in children/young adults.

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of LMP; papillary; chromophobe; other oncocytic; RCC NOS
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; FH-deficient; SDH-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/SCC; sarcomas; nephroblastic; neuroendocrine; lymphoma; epithelioid AML ; metastases

Trap: epithelioid AML = malignant; classic AML, papillary adenoma, oncocytoma = benign. Type 1/2 PRCC split **abandoned** (old type 2 = MET-altered + MiT translocation + FH-deficient).

2 · Pathology & genetics

Common and important subtypes

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Golden-yellow, hypervascular; clear cytoplasm (glycogen, lipid); CA-IX+; sarcomatoid 5–8%	Most venous invasion; worse localised, better metastatic (TKI/ICI responsive)
Papillary (10–15%)	Trisomy 7, 17 , loss Y; MET altered 81%	Hypovascular; papillae on fibrovascular cores; CK7, AMACR; multicentric (~40%)	Good (5-y CSS 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17 (hypodiploid); TP53, PTEN; FLCN in BHD	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	Intercalated cells; >90% cancer-free 5 y unless sarcomatoid
Clear cell papillary tumour	—	Linear papillae; CK7; cup-like CA-IX	ESRD, VHL; now considered benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8+; diagnosis of exclusion	Poor; may respond to cisplatin/gemcitabine

Subtype (share)	Genetics	Hallmarks	Behaviour
SMARCB1-deficient medullary	Sickle cell trait	Young Black pts; infiltrative; SMARCB1 loss	Dismal, median 8–13 mo
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss, 2SC+	Metastasises early, very poor
MIT translocation	Xp11.2 TFE3, TFEB	Clear cells + papillae; FISH diagnostic	40% of paediatric RCC; t(X;1) late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test before calling "unclassified"	Responds to ALK inhibitors
Unclassified (1–5%)	—	Catch-all, shrinking	Poor

- **Indolent group:** multilocular cystic LMP (almost uniformly benign), tubulocystic, MTSCC, eosinophilic solid & cystic (females only), ELOC, clear cell papillary, acquired cystic disease-associated.
- **Lethal group:** medullary, FH-deficient, collecting duct, unclassified, sarcomatoid.
- **Sarcomatoid** (1–5%): dedifferentiation of any subtype, not separate; = grade 4; median survival <1 y. **Rhabdoid** also mandates grade 4. Both must be reported in every RCC.
- Gross: pseudocapsule (not true capsule); cystic degeneration 10–25% (better prognosis); calcification ~10%. Venous involvement ~10%. Bilateral 2–4% sporadic; multicentric 10–20% (satellites → main cause of ipsilateral recurrence after PN).
- Papillary adenoma ≤15 mm, low grade — indolent. "Oncocytic renal neoplasm of low malignant potential, NOS" = 2022 management category.

WHO/ISUP grading (replaced Fuhrman) — clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at ×100
4	Extreme pleomorphism, giant cells, rhabdoid/sarcomatoid

Molecular biology — VHL-HIF-VEGF

Normoxia: VHL + elongin B/C + cullin-2 + RBX1 = E3 ubiquitin ligase → HIF-1α/2α degraded
VHL loss / hypoxia → **HIF-2α** accumulates → dimerises with ARNT → ↑ VEGF (angiogenesis), PDGF, TGF-α, GLUT1, EPO, CA-IX → hypervascular ccRCC

Target	Drugs
VEGF ligand	Bevacizumab (first VHL-pathway agent)
VEGFR/PDGFR (TKI)	Sunitinib, sorafenib, pazopanib, axitinib, cabozantinib (also MET, RET, AXL), lenvatinib, tivozanib
mTOR (↓ HIF translation)	Temsirolimus, everolimus
HIF-2α (blocks HIF-2α-ARNT)	Belzutifan — FDA 2021 for VHL; ~50% response
Checkpoints	CTLA-4: ipilimumab; PD-1: nivolumab, pembrolizumab; PD-L1: atezolizumab, avelumab

- Knudson two-hit: familial = germline first hit → early, multifocal, bilateral. All familial RCC syndromes **autosomal dominant**; all tumour suppressor loss except **MET (oncogene)**.
- Chemoresistance (response 5–7%) from MDR proteins. Immunogenic (spontaneous regression; IL-2 durable CR 3–5%, abandoned). Immune AEs of ICI: colitis, hepatitis, dermatitis, pneumonitis.
- TCGA: VHL commonest somatic mutation (30–40%). Poor-prognosis in all subtypes: **CDKN2A** loss, **CIMP** hypermethylation, Th2 signature. BAP1 (up to 14% sporadic ccRCC) → poor, not for AS.

Hereditary RCC syndromes

Syndrome	Gene (chr)	Renal tumour	Other features	Renal management
VHL	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	CNS/retinal haemangioblastoma, phaeo, pancreatic & renal cysts, endolymphatic sac tumour, epididymal cystadenoma	AS <3 cm; NSS ≥3 cm; belzutifan
HPRC	MET (7q31) oncogene	Multiple bilateral (type 1-like) PRCC	None	AS <3 cm; NSS ≥3 cm
HLRCC	FH (1q42-43)	Aggressive, often solitary FH-deficient RCC; CDC	Cutaneous & uterine leiomyomas; uterine LMS	Prompt excision any size, wide margins ; PN only if wide margin feasible
SDH-RCC	SDHA/B/C/D, SDHAF2	Variable, early onset ≤40 y	Paraganglioma/phaeo, GIST, thyroid	Excision, PN preferred
BHD	FLCN (17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax	AS <3 cm; NSS ≥3 cm
Cowden	PTEN (10q23)	Papillary mostly (lifetime RCC 34%)	Breast (50%), thyroid	AS <3 cm; NSS ≥3 cm; screen CT/MRI
TSC	TSC1 (9q34)/TSC2 (16p13)	Multiple AMLs; ccRCC 1-3%	Cardiac rhabdomyoma, angiofibromas, LAM, epilepsy	AML: surveil <3, everolimus 3-5, embolise/excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade ccRCC	Uveal/cutaneous melanoma, mesothelioma	Excision, NSS
MITF	MITF p.E318K	RCC (>5×)	Melanoma	—

- **VHL genotype-phenotype:** type 1 (deletion/truncation) = no phaeo; type 2 (missense) = phaeo — 2A low RCC, 2B high RCC, 2C phaeo only.
- VHL: 1 in 36,000; penetrance 70-87%; RCC now commonest cause of death; RCC mean onset ~39 y, retinal angioma/phaeo earliest.
- Pathway logic: HIF/VEGF (VHL), MET (HPRC), Krebs cycle/Warburg (FH, SDH), mTOR (TSC, BHD, PTEN).

3 · Diagnosis

Presentation

- Incidental commonest. **Classic triad** (flank pain, gross haematuria, mass) rare = advanced ("too late triad").
- IVC involvement: bilateral leg oedema, **non-reducing/right-sided varicocele**, dilated abdominal veins, PE, non-functioning kidney.
- Unexplained spontaneous perirenal haematoma: >50% occult tumour (AML or RCC) → repeat CT in few months.

Paraneoplastic syndromes (10-20%)

Syndrome	Frequency / mechanism	Key point
Raised ESR	Commonest	Advanced disease marker
Hypercalcaemia	Up to 13%; PTHrP or osteolytic mets	Only one managed medically: hydration → furosemide; zoledronic acid 4 mg IV q4wk (withhold if renal insufficiency); denosumab if refractory; steroids, calcitonin; definitive = nephrectomy ± metastasectomy
Hypertension	Renin, artery compression, AV fistula	—
Polycythaemia	Erythropoietin	—

Syndrome	Frequency / mechanism	Key point
Stauffer syndrome	3-20%; IL-6	Non-metastatic hepatic dysfunction (↑ ALP almost all); exclude liver mets; normalises after nephrectomy in 60-70%; persistence = viable tumour, poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, ataxia

Imaging

- **Multiphase CT** (pre + post contrast) = single most important test: enhancement **>15-20 HU** = RCC-suspicious; fat **<-20 HU** = AML. MRI: enhancement >20%; best for IVC thrombus, masses <2 cm; macrocyclic gadolinium safe even on dialysis.
- USG simple cyst criteria: smooth wall, round, anechoic, posterior enhancement; anything else → cross-sectional imaging. CEUS if CT/MRI contraindicated.
- FDG-PET: limited (renal background uptake); not recommended for staging. **Sestamibi SPECT/CT**: mitochondria-rich **oncocytoma**/hybrid tumours take up. **¹²⁴I-girentuximab PET** targets CA-IX (ccRCC).
- Enhancement: ccRCC & oncocytoma strong; papillary/chromophobe moderate (papillary hypoenhancing). Infiltrative → lymphoma, UC, sarcomatoid, CDC, medullary, XGP.
- Complexity scores: **R.E.N.A.L.** (Radius, Endophytic/exophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines), PADUA, C-index → PN vs RN, approach, complications.

Simple cyst → observe

Complex/solid: eGFR >45 → renal CT or MRI ± contrast · eGFR 30-45 → hydrate, same · eGFR <30 → non-contrast CT or gadolinium MRI · contrast allergy → USG/CEUS or MRI

Fat → AML · complex cyst → Bosniak · solid enhancing → options ± RMB

Bosniak

Full v2019 table & management → [Ch 136](#). Malignancy: IIF 3-10% · III ~50% · IV 75-90%.

- Key discriminator: unequivocal enhancement (absent in II/IIF). Avoid TA and RMB in cystic lesions (spillage, low yield). Cystic RCC indolent → surgery for III/IV no longer mandatory.

Renal mass biopsy

- Utility-based: when result may change management; suspected lymphoma, metastasis, infection, inflammation. Not needed for young fit patients who'll have surgery anyway, or frail patients managed conservatively. **Multiple cores > FNA**.
- Safe: haematoma ~5%, transfusion <0.5%, seeding almost nil. Specificity 94%, PPV 98.8%; **non-diagnostic ~14%** (↓ with repeat); subtype accuracy ≥90%, grade 60-80%. Non-malignant result ≠ benign.

TNM (AJCC 8th)

T	Definition	5-y CSS
T1a / T1b	≤4 / >4-7 cm, confined	95-100 / 85-95%
T2a / T2b	>7-10 / >10 cm	65-85 / 55-75%
T3a	Renal vein/segmental branches, pelvicalyceal system (8th ed.) , perirenal/sinus fat, within Gerota	45-75%
T3b	IVC below diaphragm	40-55%
T3c	IVC above diaphragm or caval wall invasion	20-45%
T4	Beyond Gerota, incl. contiguous ipsilateral adrenal	0-35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0-30%

Stage I T1N0 · II T2N0 · III T1-2N1 or T3 any N · IV T4 or M1.

- Edition changes: 6th split T1 at 4 cm; 7th split T2 at 10 cm, adrenal contiguous → T4, renal vein → T3a, N1/N2 merged; 8th pelvicalyceal → T3a.

- Staging work-up: abdominal CT + CXR (chest CT if symptoms/high risk); bone scan only if ↑ALP, bone pain, poor PS; no PET. Nodes ≥2 cm almost always malignant; smaller often reactive. CT sensitivity for thrombus: renal vein 78%, IVC 96% (false negatives right side). MRI premier for IVC and organ invasion.

Prognosis

- Pathological stage** = single most important factor; then grade, size, subtype, necrosis, sarcomatoid. Big drop with T4, N1, M1 (node+ 5-y 5–30%).
- Venous thrombus with otherwise confined disease: 5-y 45–69% — high stage but curable. Wall invasion as important as level.
- Adverse clinical: symptoms, weight loss >10%, poor PS, anaemia, hypercalcaemia, ↑ESR, ↑LDH/ALP, thrombocytosis.
- Nomograms (e.g. Kattan) outperform stage but must be revalidated. ASSURE post hoc: recurrence algorithms barely beat stage + grade → AUA risk groups use stage/grade. **Beyond notes (classic viva): UISS** = TNM + grade + ECOG PS; **SSIGN** (Mayo) = Stage, Size, Grade, Necrosis; **Leibovich** = pT, pN, size, grade, necrosis → recurrence risk in ccRCC.

Screening

- No population screening (low incidence, ~80% of SRMs benign/indolent; dipstick useless). Not in ADPKD.
- ESRD (acquired cystic disease in 80%; RCC 1–2% of these; RR 5–20×): USG/CT from 3rd dialysis year if fit. Continue native kidney imaging post-transplant.
- VHL: exam/eye annually from infancy; metanephrines from 5; CNS MRI + audiogram 2-yearly from 11; **abdominal MRI 2-yearly from 15**.
- Genetic counselling: RCC at ≤46 y, multifocal/bilateral, suggestive personal/family history or pathology.

4 · Management

Localised RCC — AUA 2021 principles

- Evaluate: multiphase imaging, CMP, CBC, urinalysis, chest imaging; assign **CKD stage (eGFR + proteinuria)**. Nephrology referral: GFR <45, proteinuria, diabetic CKD, or expected post-op GFR <30.
- PN prioritised for cT1a**; nephron-sparing prioritised for solitary kidney, bilateral, familial, CKD/proteinuria.
- RN preferred only if all 3**: (1) high complexity, PN hard even for experts; (2) no CKD/proteinuria; (3) normal contralateral kidney and new baseline GFR likely >45.
- TA: alternative for cT1a <3 cm; percutaneous; RMB before preferred; RFA = cryo.
- LND for clinically positive nodes; adrenalectomy only if imaging/operative suspicion; MIS when not compromising outcome.

Function and CKD

- ~25% of RCC patients have CKD pre-op; CKD after RN 65% vs PN 20%.
- CKD-surgical (one-time insult, ~0.7%/y decline, survival ≈ normal if GFR >45) vs CKD-medical (progressive ~4.7%/y, ↓ survival).
- PN preserves ~90% global function; RN 65–70%; compensatory hypertrophy 20–30%.
- New GFR after RN ≈ global GFR × contralateral SRF × 1.25** (SRF by parenchymal volume analysis).

Radical nephrectomy

- Robson: early artery → vein ligation, outside Gerota, adrenalectomy, extended LND. Today: perifascial dissection essential (~25% of cT1b/T2 invade fat); **no routine adrenalectomy** or routine extended LND. Laparoscopic RN standard for non-invasive tumours. Technique → Ch 141/142.

Partial nephrectomy

Indication	Situations
Imperative	Bilateral RCC, solitary functioning kidney
Relative	Contralateral kidney threatened (RAS, stones, diabetes, HTN, hydronephrosis)
Elective	cT1a, normal contralateral kidney (standard of care)

- Steps: pedicle occlusion → excision with rim → collecting system closure/vessel ligation → capsular reconstruction. Negative margin width immaterial.
- Function determinants: pre-op parenchymal quality (ceiling) > **quantity preserved (primary, modifiable)** > ischaemia. Hypothermia if ischaemia ≥ 25 –30 min. Off-clamp no advantage over limited warm ischaemia in RCTs.
- Solitary kidney: ≥ 20 –30% of one kidney needed to avoid ESRD; hyperfiltration (>50% loss) → proteinuria, FSGS → ACEi + low-protein diet.
- Bilateral: PN side first, then RN if needed. Non-feasible imperative PN → neoadjuvant TKI (stop several half-lives peri-op).
- Elective PN local recurrence 1–2% (mostly occult multifocal, distant from bed). Commoner urine leak/bleeding than RN.
- **Enucleation** (blunt along pseudocapsule): established in familial RCC; controversial in sporadic (40–50% infiltrate pseudocapsule). Report via SIB protocol.

Thermal ablation → Ch 143

- Best <2.5–3 cm; higher local recurrence than PN (salvaged by repeat TA; no CSS difference). Cryo lethal ≤ -20 °C, 2 freeze-thaw cycles, iceball 1 cm beyond tumour. Central/nodular enhancement in ablation bed = recurrence.

Active surveillance

- SRM growth 0.09–0.34 cm/y; metastasis <2% over 2–5 y. **DISSRM** registry supports AS.
- Favouring AS: elderly, life expectancy <5 y, comorbid, frail; tumour <3 cm, growth <5 mm/y, cystic, low complexity. Strong consideration for solid <2 cm or predominantly cystic.
- Imaging every 3–6 mo (AS) or 6–12 mo (expectant). **Triggers**: size >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms.

Familial RCC — the 3-cm rule (NCI)

- VHL solid tumours <3 cm: no metastases at ~5 y; ~25% metastasise if larger → intervene (PN/enucleation/TA) at **3 cm**. Applies to VHL, HPRC, BHD. **HLRCC exception** — treat at any size.
- Recurrence near-universal → lifelong surveillance; belzutifan reduces interventions (renal response 30–50%; anaemia, fatigue). Exclude phaeo/CNS haemangioblastoma before surgery.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp; keep $\geq 50\%$ lumen
II	Below hepatic veins	T3b	Sequential clamp: infrarenal IVC → contralateral RV → suprarenal IVC; lumbar
III	Intrahepatic, below diaphragm	T3b	Liver mobilisation, milk down; Pringle; venovenous bypass often
IV	Above diaphragm	T3c	CPB $\pm$ hypothermic circulatory arrest

- Imaging close to surgery (MRI or CT); TOE intra-op. **No routine pre-op embolisation** (↑ complications/mortality). Coronary angiography before CPB (levels III–IV).
- Bland distal thrombus → remove, or ligate IVC/filter. Level IV mortality 5–10%.

Locally invasive (T4) and nodes

- Only ~40% of suspected invasion confirmed; en bloc resection only cure; debulking rarely (1-y survival 10–20%); neoadjuvant therapy as "litmus test".
- Only 30–43% of enlarged nodes malignant. Node risk factors (Blute): ≥ 10 cm, cT3/T4, grade 3–4, sarcomatoid, necrosis; ≥ 2 factors → >40% node+ (prospective). 45% of positive nodes away from hilum.
- Bottom line: LND for clinically suspicious nodes, mainly staging.

Local recurrence

- After RN 2–4% (only 20–40% isolated → full work-up; resection → 30–50% long-term cancer-free). After PN 1.4–10% (repeat PN, completion, TA, AS). Positive PN margin → surveillance; completion only if gross.

Adjuvant / neoadjuvant

- Failed adjuvant: IL-2, IFN- α , MPA, RT, vaccines, girentuximab, most TKIs.
- Neoadjuvant TKI: 10–30% shrinkage; may enable PN/thrombectomy. Metastatic systemic therapy → [Ch 144](#).

Follow-up (AUA risk groups)

Risk	Definition	Chest
Low	pT1 G1-2	CXR
Intermediate	pT1 G3-4 or pT2	CXR
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross +margin	CT

- Microscopic +margin → up one level. Creatinine, eGFR, proteinuria every visit. USG may alternate after 2 y (LR/IR). **30% of recurrences after 5 y**. Post-TA: contrast imaging within 6 mo, then IR protocol.

Other renal malignancies

Tumour	Clues	Management
Sarcoma (1–2%)	Leiomyosarcoma commonest (50–60%) then liposarcoma; capsule/sinus origin, huge, no nodes, hypovascular (except haemangiopericytoma); osteosarcoma mimics staghorn	Wide en bloc excision, negative margins (first resection best chance); margin + grade = prognosis
Lymphoma	NHL; multiple masses (45%) commonest; massive/atypical adenopathy, splenomegaly	Biopsy, avoid nephrectomy ; R-CHOP $\pm$ RT
Metastases	Commonest renal malignancy (12% at autopsy); lung commonest; multifocal, 5–30 HU	Biopsy; systemic/palliation
Carcinoid	Horseshoe kidney (82$\times$)	Excision; good prognosis
Small cell	Advanced	Surgery + platinum
PNET	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than children

Non-RCC malignancies are almost all **hypovascular**; lymphoma and metastasis are biopsied, not resected. A solitary enhancing mass after organ-confined non-renal cancer is more likely RCC.

5 • Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	RCT RN vs elective PN, ≤ 5 cm, normal contralateral	PN better function; 10-y OS better with RN (81 vs 76%)	Flawed; PN functional gain may be smaller than believed
EORTC 30881	RCT RN $\pm$ LND	No survival benefit	Underpowered, low-risk (4% N+); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk ccRCC	DFS 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS HR 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry AS vs intervention, SRM	1 death/metastasis on AS; 24% delayed intervention	AS safe for SRM

6 · Theory answer frames

Q: Hereditary renal cell carcinoma syndromes.

1. **Intro** – 4-6%; AD; Knudson two-hit; early, bilateral, multifocal.
2. **Table** – gene/chromosome/tumour/features (see syndrome table).
3. **VHL in depth** – pathway, genotype-phenotype types 1/2A-C, screening protocol.
4. **Management** – 3-cm rule, NSS/enucleation, HLRCC exception, belzutifan, lifelong surveillance.
5. **When to suspect** – age ≤ 46 , bilateral/multifocal, family history → counselling.

Q: Management of a 3.5-cm incidental enhancing renal mass.

1. **Risk** – 20% benign, 60% indolent, 20% aggressive; male, size.
2. **Evaluation** – multiphase CT/MRI, chest imaging, eGFR + proteinuria, R.E.N.A.L.
3. **RMB** – utility-based indications, accuracy, limits.
4. **Options** – AS (criteria, triggers), PN (preferred), TA (<3 cm), RN (3 AUA criteria).
5. **Function** – CKD-S vs CKD-M, EORTC 30904.
6. **Follow-up** – AUA risk groups.

Q: RCC with IVC tumour thrombus.

1. **Incidence/clinical** – 4-10%; oedema, varicocele, PE.
2. **Imaging** – MRI/CT close to surgery, TOE; no routine embolisation.
3. **Levels I-IV with T stage and control technique** (table).
4. **Prognosis** – cure 45-70%; wall invasion, nodes, mets.
5. **Adjuncts** – bypass, neoadjuvant therapy, adjuvant pembrolizumab.

Q: Paraneoplastic syndromes of RCC.

1. **Basis** – 10-20%; kidney's endocrine products (EPO, renin, vitamin D, PTHrP).
2. **Table** – ESR, hypercalcaemia, HTN, polycythaemia, Stauffer, others.
3. **Hypercalcaemia management** – only medically treatable.
4. **Significance** – advanced disease; resolve after nephrectomy; persistence = residual tumour.

Q: WHO 2022 classification and molecular biology of RCC.

1. **Principle** – subtype = nephron segment + genetic lesion + behaviour.
2. **Common subtypes** – table; ISUP grading.
3. **Molecularly defined group** – FH, SDH, TFE3/TFEB, ALK, ELOC, SMARCB1.
4. **VHL-HIF-VEGF** – diagram and drug targets; mTOR, MET, checkpoints.
5. **Prognostic genomics** – CDKN2A, CIMP, BAP1.

7 · Viva corner

Q. Why does localised ccRCC do worse but metastatic ccRCC better than papillary?

→ ccRCC is more aggressive, but VHL-driven angiogenesis and immunogenicity make it respond to TKIs and ICIs.

Q. Why is HLRCC the exception to the 3-cm rule?

→ FH-deficient tumours metastasise early even when small — excise promptly with wide margins.

Q. Which paraneoplastic syndrome can be treated medically?

→ Hypercalcaemia — hydration, furosemide, zoledronic acid, denosumab.

Q. What is Stauffer syndrome and its significance?

→ Non-metastatic hepatic dysfunction (IL-6); failure to normalise after nephrectomy = residual tumour, poor prognosis.

Q. When is RN preferred over PN (AUA)?

→ All three: high complexity, no CKD/proteinuria, normal contralateral kidney with expected GFR >45.

Q. How do you predict GFR after RN?

→ Global GFR × contralateral split function (volumetry) × 1.25.

Q. Why is surgically induced CKD less harmful?

→ One-time insult, stable (~0.7%/y), survival near normal if GFR >45; medical CKD is progressive.

Q. A right-sided varicocele that does not reduce on lying?

→ Suspect renal mass with renal vein/IVC thrombus.

Q. Is routine adrenalectomy/LND needed at RN?

→ No — only for imaging/operative suspicion; EORTC 30881 showed no benefit of LND.

Q. Renal mass in a patient with massive retroperitoneal adenopathy and splenomegaly?

→ Suspect lymphoma — biopsy, avoid nephrectomy, R-CHOP.

Q. What does the grade 4 ISUP category include?

→ Extreme pleomorphism, giant cells, rhabdoid or sarcomatoid differentiation.

Q. Why avoid biopsy/ablation in Bosniak III-IV?

→ Poor yield, sampling error and risk of tumour spillage.

Q. Should you embolise before IVC thrombectomy?

→ Not routinely — no benefit, more complications and mortality.

Q. Young Black patient with sickle trait and infiltrative medullary mass?

→ SMARCB1-deficient medullary carcinoma — dismal, platinum chemotherapy.

8 · MCQ one-liners

1. Most accepted environmental risk factor: **tobacco**.
2. Strongest predictors of malignancy in a renal mass: male sex and size.
3. ~40% of masses ≤2 cm are benign.
4. VHL gene: 3p25; binds elongin B/C; degrades HIF-α.
5. Main HIF isoform driving ccRCC: **HIF-2α**; inhibitor belzutifan.
6. VHL type 2C: pheochromocytoma only.
7. Only oncogene familial RCC: **MET (HPRC, 7q31)**.
8. BHD: FLCN 17p11.2 — chromophobe/hybrid oncocytic tumours, pneumothorax.
9. HLRCC: FH 1q42; uterine/cutaneous leiomyomas; solitary aggressive tumour.
10. Papillary RCC: trisomy 7 and 17; ccRCC: 3p loss, 5q gain.
11. Chromophobe: losses of 1, 2, 6, 10, 13, 17; "plant cells"; perinuclear halo.
12. Cup-like CA-IX, CK7+: clear cell papillary tumour (benign).
13. Medullary carcinoma: sickle cell trait, SMARCB1 loss.
14. Commonest paediatric RCC: Xp11 TFE3 translocation.
15. Commonest paraneoplastic finding: raised ESR.
16. Hypercalcaemia in RCC: up to 13%, PTHrP.
17. Fat <-20 HU on CT = AML.
18. Enhancement >15-20 HU = suspicious for RCC.
19. Sestamibi uptake: oncocytoma (mitochondria).
20. Bosniak III ~50%, IV 75-90%, IIF 3-10% malignant.
21. RMB non-diagnostic ~14%; PPV 98.8%.
22. TNM 8th: pelvicalyceal invasion = T3a.
23. Contiguous adrenal = T4; non-contiguous = M1.
24. Caval wall invasion = T3c at any level.
25. Renal vein CT sensitivity 78%; IVC 96%.
26. Nodes ≥2 cm on CT almost always malignant.
27. Genetic counselling for RCC at age ≤46.
28. VHL abdominal MRI from age 15, every 2 years.
29. ADPKD: no routine RCC screening.
30. Remnant ≥20-30% of one kidney avoids ESRD.
31. Hypothermia if ischaemia ≥25-30 min.
32. Compensatory hypertrophy after RN: 20-30%.
33. 30% of recurrences after 5 years.
34. Adjuvant pembrolizumab: KEYNOTE-564, DFS HR 0.63.
35. Commonest renal sarcoma: leiomyosarcoma.
36. Commonest malignant renal neoplasm overall: metastasis (lung primary).
37. Renal carcinoid: horseshoe kidney, 82× risk.
38. Renal PNET: t(11;22), CD99.
39. Commonest CT pattern of renal lymphoma: multiple masses.

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

Reading time ≈ 27 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★★

1 · Core concepts & epidemiology

- UTUC = **5-10%** of urothelial cancers; incidence ≈ 2/100,000/yr, rising (ureteric tumours, better diagnosis, longer survival after bladder cancer).
- Peak age 70-90; M:F ≈ **2:1**; Balkans: UTUC = **40%** of renal neoplasms.
- Pelvic/ureteric ≈ 2× ureteric. Bilateral ≤ 5%; multifocal **10-20%**; concomitant CIS 11-36%; concurrent bladder cancer **17%**.
- At diagnosis ≈ **60% invasive**, 7-9% metastatic.
- After UTUC: bladder recurrence **22-47%**; contralateral upper tract 2-6%.
- 5-yr DSS: in situ 95%, localised 89%, regional 63%, distant 17%.

2 · Risk factors, pathology & genetics

Risk factors

- Smoking** – most important modifiable; RR **2.5-7**; aromatic amines → N-hydroxylamine; dose-response; ureter > pelvis; cessation > 10 yr ↓ risk 60-70%.
- Occupational "amino tumours"** – benzidine, β-naphthylamine (dyes, rubber, textiles, petrochemicals, coal); exposure ≈ 7 yr, latency up to 20 yr.
- Iatrogenic** – cyclophosphamide (acrolein), anthranoid laxatives > 1 yr, pelvic RT.
- Chronic infection/stones/obstruction** – RR 1.5-2; → SCC (less often adenocarcinoma).

Feature	Aristolochic acid (BEN, Chinese herb)	Arsenic (blackfoot)	Analgesic (phenacetin)	Lynch syndrome
Setting	Rural Balkans; Belgium 1992-93 slimming herbs; Taiwan/China	SW Taiwan, Chile well water	Chronic abusers	MMR mutation families
Mechanism	Aristolactam-dA adducts → A:T→T:A transversion (TP53 codon 139)	Arsenic; causal link unproven for UTUC	Papillary necrosis; capillarosclerosis	MMR deficiency (MSH2 highest)
Sex / age	No sex bias; ≈10 yr earlier	F 2:1; 55-60	Latency ≈ 22 yr	< 60 (mean 55); more women
Site	Bilateral more often	Ureter 2:1 over pelvis	Pelvis > ureter; think contralateral	Upper tract > bladder
Inherited?	Familial, not inherited (shared diet)	No	No	Yes, AD

All four: multiple, bilateral recurrence (field disease). Aristolochic acid: < 10% of exposed develop UTUC; bladder cancer not ↑ in BEN families. Analgesic abuse history in 22% pelvic, 11% ureteric tumours.

Lynch syndrome

- MLH1, MSH2, MSH6, PMS2; UTUC risk ≈ **14x**, **75x** in MSH2 carriers. Germline MMR mutation in **5-9%** screened UTUC.
- Suspect if (EAU): age < 60; personal LS-spectrum cancer; FDR < 50 yr with LS cancer; two FDRs with LS cancer → germline sequencing → screen colorectal/endometrial/ovarian etc. → counselling. Abnormal MMR IHC in sporadic-looking tumour → germline testing.

Molecular

- UTUC and bladder cancer genetically distinct; in same patient usually clonally related.
- 5 subtypes: hypermutated, TP53/MDM2, **RAS (worst)**, FGFR3, triple-negative. Higher stage → more TP53/MDM2.

- Key differences from bladder: **MSI** (independent prognostic; flags LS; inverted growth) and hypermethylation.
- Markers adverse almost uniformly (Ki-67, HIF-1α, Snail, E-cadherin loss, MMPs); favourable when expressed: **uroplakin III, p27**; RON no correlation. None in routine use.

Histopathology

- > 90% urothelial; variants ≈ **25%** (squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial). **Micropapillary** = independent predictor of worse PFS/CSS.
- Progression hyperplasia → dysplasia → CIS. CIS may appear as whitish plaque or red patch.
- Inverted papilloma: malignancy ≈ **18%** (ureteric) → follow ≥ 2 yr.
- SCC 0.7–7%, 6× commoner in pelvis, invasive at presentation; adenocarcinoma < 1%, calculi/obstruction, poor prognosis.

3 · Diagnosis & staging

- **Haematuria** commonest (56–98%); flank pain 20–32% (dull; colic with clots); mass ≈ 10%; incidental ≈ 15%. Systemic symptoms → metastatic work-up.
- Ureteric tumours: **70% distal**, 25% mid, 5% proximal. 85% papillary; invasion in > 80% sessile vs 50% papillary.
- **Cystoscopy always**. +ve cytology with normal cystoscopy/urethra → evaluate upper tracts.

Test	Key points
CT urography (first-line; EAU preferred to MRU)	Sens ≈ 92% , spec ≈ 95%. Filling defect (50–75%), obstruction, non-visualisation. Misses flat lesions; poor for invasion of small lesions. UC ≈ 46 HU (10–70) vs uric acid stone ≈ 100 HU.
Hydronephrosis	Invasive ureteric tumour in up to 80%; independent adverse predictor.
MR urography	If CT contrast/radiation contraindicated; sens ≈ 75% for < 2 cm; avoid gadolinium if CrCl < 30.
Retrograde pyelography	Accuracy ≈ 75%; with URS.
FDG-PET/CT	Nodes/metastases when iodinated contrast contraindicated.
Renal scan	Split function; contralateral assessment mandatory.
Flexible URS + biopsy	Accuracy 75–85% → ≈ 90%. Biopsy grades ≈ 90%; visual grading ≈ 70%. Understages (Ta biopsy often T3 at RNU); G3 biopsy predicts stage > 90%.
Cytology	Sens ≈ 65%: G1 20%, G2 45%, G3 75%. Selective ureteric washing better; brush + cytology ≈ 90%. Collect before contrast .
FISH / NMP22	FISH more sensitive than cytology; best in low-grade surveillance; not routine.

- URS indications: diagnostic uncertainty, kidney-sparing planned, solitary kidney. **URS before RNU ↑ intravesical recurrence** (esp. pelvic tumours, biopsy), no survival effect.
- Antegrade access: low-risk pelvic tumour inaccessible to flexible URS; tract seeding risk.
- EAU: stage cannot be assessed from URS biopsy → stratify by grade + appearance + hydronephrosis + cytology.

TNM (8th ed.) & grading

T	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or renal parenchyma ; ureter: periureteric fat (pT3a microscopic parenchymal, pT3b macroscopic/peripelvic fat – worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single ≤ 2 cm / single > 2 cm or multiple

- AJCC: 0a/0is, I = T1, II = T2, III = T3, **IV = T4 or any N+ or M1.**
- Nodes: pelvis/upper ureter → hilar, para-aortic, paracaval, interaortocaval; lower ureter → common iliac/pelvic.
- Grading: WHO 2004/2016 (PUNLMP, low, high grade) – basis of EAU; PUNLMP managed as low grade.

Risk stratification (EAU 2020)

Low risk — ALL required	High risk — ANY ONE
Unifocal	Multifocal
< 2 cm	> 2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy
—	Variant histology

Trap: small, low-grade unifocal tumour **with hydronephrosis = high risk.**

4 · Prognosis, recurrence & management overview

Prognostic factors

- **Stage = single most important**, then grade. SEER 5-yr CSS: ≤ T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- Stage-for-stage UTUC ≈ bladder for pT2–3; NMIBC reaching cystectomy does worse (self-selected aggressive); pT4 UTUC worse.
- **Ureteric/multifocal worse than pelvic** — renal parenchyma is a barrier; periureteric plexus rich in vessels/lymphatics.
- Preop: age (but not a bar to RNU), smoking, ASA/ECOG, obesity, delay (**RNU within 12 wk**), blood markers (low albumin, high NLR, CRP, De Ritis ratio).
- Postop: nodes (up to 40%), extranodal extension, LVI (≈ 20%; independent in N0/Nx; not on biopsy), positive margins, necrosis > 10%, sessile, concomitant CIS, variant histology, LN density ≥ 30%.
- Size: **2 cm** best cut-off for predicting ≥ pT2.
- Metastases mostly within 2–3 yr: lung, liver, bone, nodes.

Recurrence patterns

- **Bladder after UTUC:** 15–75% at 5 yr (22–47% contemporary), usually within 2 yr — downstream seeding (monoclonal) + field effect; both operate.
- **Upper tract after bladder cancer:** 2–4%, late (median > 3 yr); after cystectomy 0.75–6.4%; ≈ **30%** after BCG for CIS / BCG-refractory cystectomy → annual upper tract imaging (CT) in high-risk bladder cancer; diversion type irrelevant.

Patient	Tumour	Treatment
Male, prior bladder cancer, smoking, CKD	+ve cytology, ureteric, multifocal, invasive pT, necrosis	Laparoscopic, extravesical cuff, +ve margins, diagnostic URS

↑ Predictors of intravesical recurrence after RNU

- **Pan-urothelial disease:** bladder + ≥ 2 extravesical sites (women: both upper tracts; men: upper tract(s) ± prostatic urethra). No algorithm: cystectomy for high-grade bladder disease, conservative for non-infiltrating UTUC; total urinary tract removal option in young. Cystectomy then RNU → loss of renal function → cisplatin ineligible.

Management overview

Cystoscopy + CTU ± URS biopsy/cytology → risk-stratify
Low risk → kidney-sparing (endoscopic / segmental) with intensive surveillance
High risk → RNU + bladder cuff (± LND) within 12 wk
cN+ → systemic chemotherapy first → surgery if response
After RNU → single postoperative intravesical chemo dose → cystoscopic follow-up for all

- Pre-RNU nomograms predict $\geq$ pT2/non-organ-confined → select for **neoadjuvant chemotherapy** (pathological downstaging ↑ OS); postop nomograms guide adjuvant chemo and follow-up. Perioperative chemo trials (POUT etc.) → [Ch 139](#).
- LND at RNU: optimal staging; curative role debated.
- Technique (RNU, distal ureter, endoscopic ablation, segmental ureterectomy, topical instillation) → [Ch 139](#).
- Follow-up: cystoscopy for all; kidney-sparing patients intensively (URS, cytology, CTU); lifelong upper tract surveillance after high-risk bladder cancer.

6 · Theory answer frames

Q: Risk factors for UTUC (short note)

1. **Genetic** – Lynch, MMR genes, MSH2 75×; EAU screening criteria.
2. **Tobacco, occupational amines** – RR, mechanism, latency.
3. **Endemic** – aristolochic acid, arsenic, analgesics (table).
4. **Others** – infection/stones (SCC), cyclophosphamide, laxatives, RT.
5. **Implication** – field disease, bilateral, affects kidney-sparing choice.

Q: Diagnostic evaluation of a suspected upper tract tumour

1. **Clinical** – haematuria, flank pain, systemic features.
2. **Cystoscopy** – always.
3. **CTU** – findings, HU, hydronephrosis, limitations; MRU/PET alternatives.
4. **Cytology** – grade-dependent; selective washing; before contrast.
5. **URS biopsy** – indications, understaging, ↑ bladder recurrence.
6. **Risk stratification** – table; LS screening.

Q: Prognostic factors in UTUC

1. **Stage/grade** – TNM, AJCC, CSS by stage.
2. **Preoperative** – location, multifocality, size, hydronephrosis, age, smoking, delay, biomarkers.
3. **Postoperative** – N+, LVI, margins, necrosis, architecture, CIS, variants.
4. **Nomograms & risk groups**.

Q: Bladder recurrence after UTUC and UTUC after bladder cancer

1. **Incidence & timing** – asymmetric.
2. **Theories** – seeding vs field effect.
3. **Predictors** – patient/tumour/treatment table.
4. **Prevention** – single intravesical chemo dose; avoid unnecessary URS.
5. **Surveillance** – cystoscopy; annual upper tract CT in high-risk bladder cancer.

7 · Viva corner

Q. Why is UTUC more often invasive than bladder cancer?

→ Much thinner (mesodermal) muscle layer allows early invasion.

Q. Why can't URS biopsy stage UTUC?

→ Small, superficial samples rarely include deep layers; Ta often upstaged to T3. Grade is reliable.

Q. Pelvic tumour invading renal parenchyma — T?

→ T3 (pT3a microscopic). Through kidney into perinephric fat = T4.

Q. Why do pT3 pelvic tumours outlive pT3 ureteric?

→ Parenchyma acts as barrier; periureteric tissue has rich lymphovascular plexus.

Q. Is BEN inherited?

→ Familial, not inherited — shared dietary aristolochic acid; signature A:T→T:A TP53 transversion.

Q. Who needs Lynch screening?

→ < 60, personal LS cancer, FDR < 50 with LS cancer, or two FDRs with LS cancer.

Q. Filling defect on CTU — how to tell tumour from uric acid stone?

→ HU: UC ≈ 46 vs uric acid ≈ 100 (80–250).

Q. When to send cytology relative to RGP?

→ Before contrast — ionic agents degrade cells.

Q. Downside of URS before RNU?

→ ↑ intravesical recurrence (esp. pelvic tumours, biopsy); survival unaffected.

Q. Small low-grade tumour with hydronephrosis?

→ High risk — one adverse feature suffices.

Q. cN+ UTUC — management?

→ Systemic chemo first; surgery if response.

Q. Why neoadjuvant rather than adjuvant thinking in UTUC?

→ RNU loses renal function → cisplatin ineligibility; nomograms select ≥ pT2.

8 · MCQ one-liners

1. Bladder recurrence after UTUC: 22–47%; contralateral 2–6%.
2. Smoking RR 2.5–7; ureter at greater risk.
3. Lynch: 14× UTUC risk; MSH2 75×.
4. Germline MMR mutation in 5–9% of screened UTUC.
5. RAS-mutated subtype: worst prognosis.
6. Aristolochic acid signature: A:T→T:A transversion, TP53 codon 139.
7. Chinese herb nephropathy: Aristolochia fangchi substituted for Stephania tetrandra.
8. Ochratoxin A: histology doesn't match BEN tumours.
9. Arsenic: blackfoot disease; women 2:1; ureter 2:1.
10. Phenacetin latency ≈ 22 yr; capillarosclerosis pathognomonic.
11. Variant histology ≈ 25%; micropapillary independently adverse.
12. Ureteric inverted papilloma: malignancy ≈ 18%; follow ≥ 2 yr.
13. CTU sensitivity 92%, specificity 95%.
14. UC ≈ 46 HU; uric acid stone ≈ 100 HU.
15. URS raises diagnostic accuracy to ≈ 90%.
16. N1 = single node ≤ 2 cm; any N+ = AJCC stage IV.
17. Size cut-off predicting ≥ pT2: 2 cm.
18. Necrosis > 10% = independent adverse predictor.
19. LN density ≥ 30% predicts poor outcome.
20. BCG for bladder CIS: ≈ 30% UTUC → annual upper tract imaging.
21. Uroplakin III and p27 expression favourable.
22. MSI: key difference from bladder cancer; flags Lynch.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Reading time ≈ 33 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★

1 · Core concepts & decision-making

Epidemiology, pathology, staging, diagnosis, risk groups, systemic therapy → [Ch 138](#)

- **Standard:** RNU + bladder cuff, if contralateral kidney normal and no surgical contraindication.
- UTUC is **unforgiving of spillage** – seeds/implants, not salvageable → every approach keeps the system closed.
- Ureteroscopic biopsy is shallow → cannot stage; stage **inferred** from grade + appearance + imaging + cytology.
- URS: accuracy ≈ 75% → 85–90%; essential if diagnosis uncertain or KSM planned.
- **Kidney-sparing (KSM):** low risk → regardless of contralateral kidney. **Imperative** (selected high risk): CKD, solitary kidney, bilateral tumours, poor surgical candidate.
- **Lower ureteric** disease (esp. unfavourable) → segmental/distal ureterectomy rather than endoscopy.

AUA/SUO group	Features	Ablation	Systemic
Low – favourable	LG, cytology -ve, unifocal, papillary, no obstruction/invasion	Preferred	No
Low – unfavourable	LG + obstruction / multifocal / prior lower-tract UC	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select	NAC/adjuvant
High – unfavourable	HG cytology, invasion, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	NAC/adjuvant

Trap: low grade ≠ automatic ablation – obstruction or multifocality → unfavourable; RNU keeps a role.

Imaging suspicion → CT/MR urogram → ureteroscopy + biopsy + cytology → risk group + shared decision
High risk + adequate GFR → RNU ± NAC · Low risk or poor GFR → conservative (endoscopic / segmental / topical)

2 · Radical nephroureterectomy

Indications

- Non-metastatic high-risk renal pelvis / proximal ureter: HG biopsy/cytology, obstruction, parenchymal invasion.
- LG with unfavourable features (obstruction, multifocal, large volume) or failed/infeasible endoscopy.
- MIS contraindications (relative): bulky tumour invading adjacent structures; extended LND post-chemo.

Open RNU – key steps

- Genitalia in field (catheter). Midline = best nodes + bladder, poor upper pole; flank incisions need 2nd Gibson/Pfannenstiel for cuff.
- **Kidney first, cuff last:** Toldt → colon medial → **clip ureter early** (limits cell migration) → hilum before manipulation → artery then vein → kidney outside Gerota.
- Hazards: right duodenum; left pancreas tail, spleen.
- **Adrenalectomy not routine** – only if involved.
- Ureter may be divided between ties/clips/stapler at tumour-free level; **never seal with vessel-sealer alone**.

Distal ureter & bladder cuff

- Remove **entire intramural ureter + orifice**. Retained stump → local recurrence **30–75%**, worse DSS/OS.
- Prior bowel diversion → excise ureteroenteric anastomosis with bowel cuff.
- Principles: orifice with small urothelial margin → contain spillage → watertight closure.

Technique	Status / reason
Extravesical (lap/robotic; open with clamp)	Acceptable; ureter under superior vesical artery; hazard – entering tunnel, contralateral orifice
Transvesical (open, anterior cystotomy)	Acceptable; 1-cm mucosal cuff; en bloc partial cystectomy if invasive intramural tumour
Transvesical "lasso"	Abandoned – spillage, recurrence
Endoscopic "pluck" (transmural TUR)	Abandoned – extravesical spillage, recurrence
Intussusception "stripping"	Abandoned – recurrence
Lap extravesical "staple"	Abandoned – +ve margins, urothelium trapped in staple line

- **Extravesical sequence:** follow ureter to tunnel → tie at detrusor → partial bladder fill → divide lateral pedicle → empty bladder → excise cuff en bloc (traction everts orifice) sparing contralateral orifice.
- MIS: bladder entered supralateral so it drains; preplaced purse-string; flexible cystoscopy can confirm complete excision + contralateral patency.
- Closure: 2 layers absorbable (mucosa, then detrusor/adventitia); Foley several days; drain optional.

Laparoscopic / robotic RNU

- Two parts: nephrectomy + proximal ureter, then distal ureter/cuff; **intact extraction** in bag.
- Extraction incision serves distal ureterectomy: Pfannenstiel/low midline if dissection below iliacs; Gibson if higher.
- Leave lateral Gerota attachments (kidney doesn't fall onto hilum); early ureteric clip; **wide perihilar margin** (lymphatic spread).
- Ureteric blood supply: **anteromedial** proximal ⅓, **medial** middle ⅓, **lateral** distal ⅓ **MCQ**

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent (even by stage)	Comparable to LNU
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer +ve margins
LND	Fewer nodes	—	Done more often, yield maintained
Caveat	—	Possibly ↑ bladder recurrence; less favoured T3/T4	Longest time; retrospective data

Access matters less than principle: port-site seeding and peritoneal carcinomatosis trace to **spillage**, not approach.

Lymphadenectomy

- Staging + therapeutic value in **T2-T4**, esp. renal pelvis/proximal ureter.
- Node +ve 12-25% overall; pTa/is 0-3% → pT4 up to 100%.
- pN0 > pNx > pN+ for outcome. **≥8 nodes** → ↑ detection, ↑ DSS (>pT1).
- Yield driven by surgeon > approach. Specific complication: **chylous/lymphatic leak**.

Primary	Right	Left	Caudal limit
Renal pelvis	Hilar, paracaval, retrocaval, interaortocaval	Hilar, para-aortic	IMA
Upper ⅔ ureter	As renal pelvis		Aortic bifurcation
Lower ⅓ ureter	Ipsilateral common/external/internal iliac, obturator, presacral		Pelvic

Drainage runs **right → left**: right tumours need interaortocaval; on left, interaortocaval adds little.

Results of RNU

- Gold standard; bladder recurrence common (drop-down implantation – genomic evidence).
- Prognostic cliff **T2 → T3**; HG survival ≈ T3; N+ behaves ≈ metastatic (5-yr OS 0-4%).
- KSM vs RNU (meta-analysis): similar OS/CSS, ~**3x reinterventions** with endoscopy.

3 · Segmental ureteral resection

- Pros: nephron-sparing + **specimen** guides adjuvant Rx.

- **Uteroscopy of entire tract above tumour mandatory** (exclude multifocality). HG → add ipsilateral pelvic LND.
- Distal > mid > proximal ureter (frequency). Ipsilateral recurrence after conservation **33-55%**, mostly distal, lifelong.
- Selected: outcomes ≈ RNU; prognosis set by stage/grade, not extent of surgery (SEER: no CSM difference).

Procedure	Indication	Key points
Segmental + uretero-ureterostomy	LG non-invasive proximal/mid tumour not endoscopically ablatable (size/number); or HG when unit must be saved	Rare. Ligate 1-2 cm above/below, frozen section margins, spatulate, 4-0 Vicryl, stent; tension-free, preserve blood supply
Distal ureterectomy + neocystostomy	LG low-stage, or selected HG/invasive distal tumour needing renal preservation	Whole distal ureter + cuff; ± LND if HG; open or robotic

- Reimplant: **non-refluxing** – confines infection, theoretically avoids upper-tract seeding; **refluxing** – easier surveillance. Debated.
- **Boari**: preop cystogram; small/irradiated bladder contraindicates; base ≥2 cm wider than apex; width ≥3× ureteric diameter; tubularise over catheter.
- **Ileal ureter**: 20-25 cm ileum, ≥15 cm from ileocaecal valve, isoperistaltic; Foley ≥1 wk; cystogram + nephrostogram before tube removal.
- After reconstruction: drain, Foley 5-7 d, cystogram if extensive.

Manoeuvre	Length gained	Condition
Psoas hitch	~5 cm	Mobile bladder; avoid nerve entrapment
Boari + hitch	10-15 cm more; may reach pelvis	Adequate capacity
Renal descensus + nephropexy	Up to 8-10 cm	Left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant – skilled hands

4 · Endourological management

- Sample then ablate (cautery/laser). **Grade is the key selector**; HG results poor.
- Large/suspected HG: **staged** – biopsy + partial ablation only, full treatment once pathology confirms.
- AUA/SUO: endoscopic Rx for **<1.5 cm**; caution ≥1.5 cm (invasion risk).
- Lifelong endoscopic + imaging commitment; recurrence expected, several sessions often needed.

	Retrograde ureteroscopic	Antegrade percutaneous
System	Closed – no non-urothelial seeding	Open – tract seeding (~0.3%, all HG)
Best for	Low-volume ureteric/small renal tumours	Large pelvis/upper ureter, lower-pole calyx, diversion
Staging	Cannot (shallow biopsy)	Can – base biopsied separately
Morbidity	Low, day case	Higher (~25%; transfusion, renal failure), admission
Recurrence (pooled)	Higher; ↑ bladder recurrence after URS	Lower
Second look/topical	Repeat URS; catheter/stent reflux	Retained nephrostomy

Uteroscopy - technique

- Cystoscopy → cytology → wire or no-touch → 6.9-7.5 Fr scope to pelvis → flexible for calyces → map → saline barbotage ↑ cytology yield.
- Tight ureter → pre-stent. Access sheath: no recommendation.
- Methods: bulk excision + base ablation (LG on stalk); resection to base (never beyond lamina propria – thin ureter); **biopsy then laser** (usual). Intramural tumour → ablation or TUR of distal ureter.
- Bugbee: unpredictable depth; **avoid circumferential fulguration** (stricture).

Laser	Penetration	Note
Ho:YAG	<0.5 mm	Ablation + haemostasis; ideal in ureter

Laser	Penetration	Note
Nd:YAG	5–6 mm	Coagulative necrosis; transmural injury risk in ureter
Thulium	<0.2 mm	1940 nm; strong coagulation

Percutaneous - technique

- Retrograde catheter → puncture: peripheral calyx – distal to tumour, never through it; pelvis/upper ureter – upper/mid calyx.
- Dilate to 30 Fr ($\pm$ 2-stage, tract matured 1–2 wk). Cold-cup, resectoscope (**bipolar** – avoids hypo-osmotic absorption) or laser; base sent separately.
- **Nephrostomy left always** → second-look nephroscopy. HG/invasive on pathology → RNU.

5 · CIS, surveillance & topical therapy

- Positive voided cytology → repeat → CTU + bladder evaluation/biopsy → if persistent after negative bladder (or post BCG "6+1") → **selective upper-tract cytology**, RPG, prostatic urethral biopsy, URS.
- Upper-tract CIS: diagnosis of exclusion; **RNU for cytology alone not recommended** (false positives, later contralateral disease). Segmental resection ineffective (field effect).
- AUA/SUO: **6-wk BCG induction** for UT-CIS; nephrostomy most reliable route; CR 41–100%, but recurrences may be lethal.

Post-RNU single-dose intravesical chemo

- Bladder cancer after RNU 30–50%, mostly within 2 yr (seeding).
- **AUA/SUO strong**: single dose after RNU and after endoscopic management (latter optional/extrapolated).
- Postop (3–7 d): cystogram confirms no leak → MMC/gemcitabine → dwell 1–2 h.
- **Intraoperative** (authors' preference): instil before incision, clamp 1–2 h, drain before cuff – no cystogram/leak cancellation.

Upper-tract instillation

Route	Method
Antegrade	Nephrostomy, gravity, ≤ 25 cm H ₂ O column
Retrograde	Ureteric catheter
Reflux	Indwelling stent + intravesical agent

- Indications: solitary kidney, bilateral UTUC, ESRD risk. Always **low pressure + no active infection** (sepsis, absorption deaths).
- Adjuvant instillation after papillary ablation: no proven benefit; complications – **sepsis, stricture**; BCG – renal granulomas (commonest), function loss if VUR.
- Adjuvant BCG acceptable after complete ablation of HG favourable Ta/T1.
- **UGN-101 (Jelmyto)**: MMC reverse-thermal hydrogel, FDA 2020, chemoablation of **LG** UTUC not endoscopically manageable. Stricture ~44%, marrow suppression, embryo-fetal toxicity; antegrade fewer strictures.

6 · Complications & follow-up

- URS: perforation (stent), stricture. Percutaneous: bleeding, absorption, UPJO. RNU: renal loss → nephrology.
- Every visit: history, exam, metabolic panel, **cystoscopy**; KSM adds ipsilateral endoscopy. RPG if contrast contraindicated.

Group	Cystoscopy	UT endoscopy	Imaging
KSM low risk	3 mo, then 6–9 mo $\times$ 2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9 mo $\times$ 2 yr, yearly to 5
KSM high risk	3 mo, then 3–6 mo $\times$ 3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6 mo $\times$ 3 yr, yearly to 5
RNU <pT2N0	+ cytology 3 mo, then per grade	—	CT/MR $\leq$ 6 mo, yearly $\geq$ 5 yr
RNU $\geq$ pT2	3 mo, 3–6 mo $\times$ 3 yr, yearly	—	A/P 3–6 mo $\times$ 2 yr; chest CT 6–12 mo $\times$ 5 yr

Perioperative systemic chemotherapy (high risk)

- **Renal function is the hinge:** CKD common and worsens after RNU → many lose cisplatin eligibility post-op → give platinum **before** the kidney goes.
- **Neoadjuvant** (AUA/SUO standard option for defined high-risk UTUC, esp. if post-op CrCl expected **<50**): downstaging, pCR, 5-yr DSS benefit (meta-analysis). EA8141: CrCl >50 → aMVAC ×4; 30–49 → gem-carbo ×4; pCR 14%, <ypT2N0 60%.
- **Adjuvant** (POUT): pT2–T4 or N+ after RNU → gem-cis (CrCl >50) / gem-carbo (30–49). Weaker with carboplatin and N+ → cisplatin preferred.

7 · Landmark trials

Trial	Design	Result	Take-home
ODMIT-C	RCT, MMC 40 mg single dose at catheter removal post-RNU	Bladder recurrence 28% → 16% at 1 yr	Single-dose intravesical chemo standard
Ito (pirarubicin)	RCT, single dose ≤5 d post-RNU; biopsy-proven	42% → 17% at 2 yr	Confirms benefit
LNU vs ONU	Only RCT; small	No 5-yr CSS/MFS difference; less blood loss, earlier discharge	MIS acceptable
UGN-101 phase III	Single-arm, LG UTUC	CR 59%; 56% disease-free at 12 mo	Gel chemoablation option; stricture
POUT	Adjuvant platinum vs surveillance, pT2–4/N+	Stopped early for efficacy; DFS HR 0.45 ; 3-yr DFS 71% vs 46%	Adjuvant platinum = standard post-RNU for ≥pT2/N+

8 · Theory answer frames

Q: Kidney-sparing management of UTUC

1. **Rationale** – nephron preservation; RNU cost = renal function.
2. **Indications** – elective low risk; imperative (solitary, bilateral, CKD, unfit) – risk table.
3. **Options** – URS vs PCN table; segmental/distal ureterectomy; topical (BCG, UGN-101).
4. **Technique pearls** – biopsy then laser, no circumferential fulguration, nephrostomy second look.
5. **Results** – similar survival, ~3× reinterventions; grade predicts recurrence.
6. **Follow-up** – lifelong; AUA/SUO schedule.

Q: Management of distal ureter and bladder cuff during RNU

1. **Why** – stump recurrence 30–75%, worse survival.
2. **Principles** – early ureteric ligation, whole intramural ureter + orifice, no spillage, watertight closure.
3. **Techniques** – extravesical vs transvesical; abandoned methods with reasons (table).
4. **MIS specifics** – supralateral cystotomy, purse-string, cystoscopic confirmation.

Q: Lymphadenectomy in UTUC

1. **Rationale** – staging, prognosis, therapeutic benefit T2–T4.
2. **Node positivity by stage.**
3. **Templates** – table; right→left drainage.
4. **Yield** – ≥8 nodes; surgeon > approach.
5. **Complications** – chyle leak.

9 · Viva corner

Q. Why clip the ureter early in RNU?

→ Prevents tumour cell migration to bladder during renal manipulation.

Q. Why not use the lap stapler across the cuff?

→ Urothelium trapped in staple line → positive margins, staple-line recurrence.

Q. Why can ureteroscopy not stage UTUC?

→ Biopsies tiny/shallow; stage inferred from grade (LG mostly Ta/T1, HG mostly invasive).

Q. Low-grade but obstructing and multifocal - ablate?

→ Low-risk unfavourable: ablation may be offered; RNU retains a role.

Q. Where do you puncture for a peripheral calyceal tumour?

→ Distal in that calyx, beyond the tumour, never through it.

Q. Why bipolar resection percutaneously?

→ Avoids absorption of hypo-osmotic irrigant.

Q. Positive cytology, normal imaging, normal URS - do RNU?

→ No; CIS diagnosis of exclusion; BCG topical, RNU only if more than surface disease.

Q. Refluxing or non-refluxing reimplant?

→ Non-refluxing confines infection/seeding; refluxing eases surveillance; debated.

Q. Intraoperative vs postoperative instillation?

→ Intraop: no cystogram, no leak cancellation, drug present when seeding occurs.

Q. Precautions for upper-tract instillation?

→ Low pressure (≤ 25 cm H₂O), sterile urine - sepsis/absorption deaths reported.

Q. Does pre-RNU ureteroscopy worsen outcome?

→ ↑ bladder recurrence; no ↑ metastasis.

10 · MCQ one-liners

1. Retained intramural stump recurrence: 30–75%.
2. AUA/SUO endoscopic cut-off: <1.5 cm.
3. Ureteric blood supply distal third: lateral.
4. Distal ureter passes under the superior vesical artery.
5. Ho:YAG penetration <0.5 mm; Nd:YAG 5–6 mm; thulium <0.2 mm.
6. Percutaneous tract seeding $\approx 0.3\%$, all high grade.
7. Psoas hitch ≈ 5 cm; Boari extra 10–15 cm.
8. Boari width $\geq 3\times$ ureteric diameter; base ≥ 2 cm wider than apex.
9. Ileal ureter: 20–25 cm, ≥ 15 cm from ileocaecal valve.
10. ≥ 8 nodes improves DSS above pT1.
11. Right renal pelvis tumour template includes interaortocaval nodes.
12. Renal pelvis template caudal limit: inferior mesenteric artery.
13. UGN-101 (Jelmyto): FDA 2020, low-grade UTUC.
14. UGN-101 stricture rate $\approx 44\%$.
15. Commonest BCG upper-tract complication: renal granulomas.
16. First ureteroscopic tumour Rx: Goodman 1984; antegrade: Tomera 1982.

141 Contemporary Open Surgery of the Kidney

Reading time ≈ 33 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★

1 · Core concepts & preoperative work-up

- Open surgery today = **exposure + early great-vessel control**: locally advanced RCC (IVC thrombus), cytoreduction, complex tumours, hostile abdomen. Minimally invasive → [Ch 142](#); RCC biology/staging → [Ch 137](#).
- All: urinalysis, SCr, eGFR (**CKD-EPI 2021, race-free** preferred over MDRD). Cystatin C independent of muscle mass/diet; more sensitive to mild GFR change.
- GFR **<60** or significant proteinuria → perioperative nephrology consult.
- Advanced RCC: CBC, metabolic panel (MSKCC/IMDC risk), LFT + coagulation (**Stauffer syndrome**), cardiopulmonary assessment.
- Imaging: CT/MRI; contrast generally OK if GFR **>30**. Suspected thrombus → **MRI (MR venography)** or CT venography, **within 1 month** of surgery (rapid propagation).
- Prophylaxis: cefazolin (or clindamycin) single dose ≤1 h before incision; no routine bowel prep; IPC before induction + UFH 5000 IU SC; stop smoking 4–6 wk; incentive spirometry.
- Renal artery embolisation**: now very selective (authors no longer use prophylactically). Postinfarction syndrome (pain, nausea, fever ≤72 h); surgery ideally same morning — dissection harder if delayed >48 h; may *increase* blood loss (oedema).

2 · Surgical anatomy & incisions

- Organs at risk: **right** — liver, duodenum, colon; **left** — spleen, pancreatic tail, colon, SMA.
- Right renal artery lies **posterior & inferior to the renal vein**, retrocaval; reachable in the **interaortocaval space** by retracting LRV cephalad.
- SMA** arises from anterior aorta **1–2 cm cephalad to LRV** — mistaken for left renal artery.
- Right gonadal & adrenal veins → IVC; left gonadal & adrenal veins → LRV.
- Cattell-Braasch** (right medial visceral rotation): white line → around caecum → Kocher → incise mesenteric root to Treitz → midgut pivots on SMA; exposes aorta/IVC.
- Direct infracolic approach** (trauma, big hilar tumour): incise posterior peritoneum below Treitz → aorta + crossing LRV.
- Left upper pole: divide splenorenal ligament; partial **Mattox** rotates spleen + pancreas medially.

Incision	Landmarks	Best for	Cons
Midline	Xiphoid → 2 cm below umbilicus	Trauma, advanced tumour/IVC thrombus (authors' choice), bilateral, renovascular	Transperitoneal, bowel mobilisation
Anterior subcostal	2 fingerbreadths below costal margin, 11th rib tip → rectus	Unilateral transperitoneal	Muscle division
Chevron	Bilateral subcostal, 11th tip → 11th tip	Renovascular; RN + IVC thrombectomy; both pedicles	Extensive muscle division
Subcostal flank	1 fingerbreadth below 12th rib	Lower pole, PUJ, proximal ureter	Poor hilar access; not for PN/large mass
Supracostal flank (11th/12th)	Upper border of rib	RN/PN; 8-cm mini-flank supra-11th	Pleura
Transcostal	On 11th rib, distal rib resected	Retroperitoneum at old rib tip	Rib division
Thoracoabdominal / dorsal lumbotomy	Largely historical (morbidity); thoracoabdominal still an option for level III–IV thrombus		

- Supine**: lumbar roll/flexion (hyperlordosis) — risks aortocaval compression, back pain; pad elbows (ulnar n.).

- **Flank:** 12th rib tip over kidney bar, axillary roll; ↓ venous return/lung function (avoid in cardiopulmonary disease, severe kyphosis); dependent-side rhabdomyolysis.
- Flank: spare **subcostal nerve** (between internal oblique & transversus); supracostal — divide intercostals on **upper border of rib**, push pleura up.

3 · Simple & radical nephrectomy

	Simple	Radical
Disease	Benign: poor function (obstruction, infection, stones, VUR, ADPKD, dysplasia); rarely functioning kidney for pain/bleeding/renovascular HTN	Tumour unsuitable for PN, nodes, cytoreduction, renal vein thrombus
Plane	Within Gerota's, on capsule (include psoas fascia if adherent; aspirate hydronephrosis)	Outside Gerota's, en bloc
Adrenal	Spared	Spared if radiologically normal

- Robson (1969): kidney + Gerota's + adrenal + nodes crus → bifurcation. Modern: spare adrenal, selective LND.
- Staging: CT/MRI + **chest CT**; no routine bone scan or brain imaging.
- Biopsy when result changes plan: benign-looking/infiltrative (UTUC, lymphoma, metastasis), before systemic therapy, bilateral tumours, trials.
- Organ invasion: altered appearance/enhancement, not contour. Liver rarely truly invaded (bilaminar capsule; <1 cm).

Right RN: survey abdomen → incise peritoneum along Toldt, iliac → hepatic flexure → Kocher (athermal) → IVC → renal vein looped/palpated for thrombus → artery behind vein → ligate **artery before vein** → kidney outside Gerota's.

Difficult hilum: interaortocaval artery ligation (0 silk) → kidney decompresses → vein doubly ligated → artery lateral to IVC divided. Or lower pole → ureter → lift kidney to show artery.

Left RN: Toldt splenic flexure → iliac → renocolic lig. (protect pancreas) → LRV on anterior aorta; follow gonadal vein to LRV → artery cranial/posterior → artery: 0 silk + 3-0 Prolene; confirm occlusion.

- **En bloc pedicle** (hilar nodes): 3 clamps, divide between 2nd and 3rd, tie ×2 + suture ligature; AVF theoretical, extremely rare.
- Lumbar veins: silk ties; **no clips on IVC side** (block clamping). Retracted lumbar → sponge-stick psoas compression, fulguration/suture.
- **Loss of pedicle control:** tell anaesthetist → resuscitate → finger/sponge compression → 2 suckers → **never clamp blindly**. Artery: compress aorta above. IVC tear: finger → Allis → repair.
- Prolene: IVC/aorta 3-0/4-0; renal vessels 4-0/5-0; artery BB needle (avoids plaque fracture).
- **Adrenalectomy** only for large upper-pole tumour, adrenal mass, diffuse involvement; adrenal mets **<5%**; CT/MRI miss 20-25%.
- **LND:** nodes positive ≈5%. Recommended when nodes enlarged (short axis **≥1.5 cm ≈ 50% positive**) — prognostic, no CSS benefit. Right template: paracaval, precaval, retrocaval, interaortocaval, crus → bifurcation; left: para-/retro-aortic, LRV → bifurcation. **Clip lymphatics above renal vein** → prevents chylous ascites.
- **Pleural check:** saline in wound + big breath → bubbles. Repair over 14-Fr catheter, evacuate on inspiration; CXR in all.

4 · IVC tumour thrombus

- RCC = commonest cause; **5-10%** of RCC. Untreated: most die within 1 yr → surgery is backbone.
- Tumour vs **bland** thrombus (coagulum below obstruction) — distinction drives management.
- Therapeutic anticoagulation if VTE, bland thrombus or (impending) IVC occlusion. **No suprarenal filters**. Embolisation rarely (Budd-Chiari unresectable, bleeding kidney).
- **TEE** for level II-IV. Cardiothoracic surgeon for III-IV, planned preop.

Mayo level	Extent	Management
0	Renal vein	RN (commonest)
I	IVC ≤ 2 cm above ostium	Milk, Satinsky partial occlusion, ostial cavotomy
II	> 2 cm, below hepatic veins	Full IVC control (infrarenal $\rightarrow$ contralateral RV $\rightarrow$ suprarenal), infrahepatic cavotomy
III	At/above major hepatic veins, below diaphragm	Liver mobilisation $\pm$ Pringle; occlusive $\rightarrow$ suprahepatic clamp; non-occlusive $\rightarrow$ VVB
IV	Above diaphragm	Sternotomy, CPB + DHCA, atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interruption (clip/staple) — not extraction
D	Mixed with tumour	Infrarenal IVC resection (preserve lumbar)

- Level I/II steps: early arterial ligation $\rightarrow$ Rummel tourniquets: **infrarenal IVC $\rightarrow$ contralateral RV $\rightarrow$ suprarenal IVC** $\rightarrow$ cavotomy $\rightarrow$ extract $\rightarrow$ heparinised saline flush $\rightarrow$ 4-0 Prolene $\rightarrow$ release **caudal first**, flush 5–10 mL (air/debris) $\rightarrow$ cranial clamp.
- Left tumour harder: both colons mobilised. Authors don't clamp right renal artery — intermittently release right RV clamp. No routine heparin.
- Level III: liver rotated (transplant technique); major hepatic veins never divided, accessory (caudate) veins ligated. **Pringle** ≤ 60 min, ideally ≤ 20 min. Budd-Chiari = contraindication.
- **Clamp trial** 2–5 min: suprahepatic clamp $\rightarrow$ preload $\downarrow 60\%$, CO $\downarrow 40\%$. Abort if **CO falls $> 50\%$ or MAP $> 30\%$** $\rightarrow$ bypass. Proceed if tolerated and extraction < 30 min.
- Bypass if: failed trial, cardiac/hepatic/contralateral renal dysfunction, portal HTN, uncontrolled bleeding.

	VVB	CPB + DHCA
Access	Femoral $\rightarrow$ IJV, percutaneous; no sternotomy	Sternotomy; cool to 16–18 °C, drain 95% blood
Heparin	Partial	Full; coagulopathy common
Drawback	Lumbar/intercostal backbleeding	Cardiac, neuro risk
Use	Least invasive; avoid if atrial thrombus, iliac bland thrombus, Budd-Chiari	Level IV; bloodless field; ≤ 60 min (< 40 better)

- **Reconstruction:** lumen narrowed to $\leq 50\%$ $\rightarrow$ primary closure; more $\rightarrow$ **patch** (bovine pericardium, smooth side in); width (Leibovich-Bower) = normal diameter $\times$ % resected $\times 3$. Circumferential $\rightarrow$ **PTFE 16–20 mm** graft; then lifelong aspirin.
- Right tumour: suprarenal IVC resectable only if LRV ligated caval to its tributaries (collateral drainage). Left tumour: right kidney has no collaterals $\rightarrow$ no suprarenal resection without alternative drainage.

"Shortcuts that are longcuts": opening IVC without full inflow control, side-biting a level I, clips on IVC side, incomplete arterial control, forceful milking, clamping before hydration/test clamp.

5 · Partial nephrectomy

- Preferred when feasible — **treatment of choice for T1**, even with normal renal function; indications now size-independent.
- Relative CI: $< 20\%$ nephron mass retained; pedicle encasement, diffuse collecting system invasion, venous thrombus, cT4, cN1.
- **Hyperfiltration:** nephron loss $> 80\%$ $\rightarrow$ FSGS, proteinuria, CKD.
- Multifocality: $\sim 2\%$ clear cell/chromophobe, $\sim 10\%$ papillary. Hereditary: observe until **3 cm** (except **HLRCC** — wide excision); IOUS for subcortical lesions.

Ischaemia issue	Evidence
Warm ischaemia	Traditional aim <30 min ; once preserved parenchyma accounted for, time matters little (within limits)
Hypothermia vs none	RCT: no functional difference (good baseline, contralateral kidney)
Mannitol vs placebo	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: shorter ischaemia, less blood loss, shorter stay

- Options: manual compression, Kaufmann parenchymal clamp, selective arterial, en bloc pedicle. Venous unclamping — no proven benefit. Always obtain hilar control first.
- Enucleation** (pseudocapsule plane, minimal normal tissue) vs **resection** (5–10 mm margin, usually needs clamping).
- IOUS: before clamping, for endophytic/central tumours.
- Renorrhaphy**: inner — 4-0 absorbable figure-of-eight to vessels, running closure of collecting system (no permanent sutures = stone nidus); outer — capsule-to-capsule over bolster, tension along suture. Unclamp → check → close Gerota's → drain.

6 · Complications

Injury	Management
SMA	Potentially lethal → vascular repair/graft
IMA / IMV	Ligation safe (Drummond marginal artery)
SMV	Never ligate; 6-0 Prolene repair/vein graft; leave abdomen open (compartment syndrome)
Liver/spleen	Argon beam, spray cautery, Surgicel, pledgeted suture; splenorrhaphy/splenectomy
Duodenum	Haematoma: expectant or drain (spare mucosa); laceration: 2-layer + omentum + drain. Don't resect "nonviable-looking" segment
Pancreas	Superficial: glue + drain (amylase/lipase); ductal → GI surgeon; fistula: drain, NPO, octreotide
Pleura	Purse-string over 14-Fr catheter; large → chest tube; CXR

- Thrombectomy**: air embolism (release caudal clamp first), PE (early arterial ligation; embolectomy via pulmonary arteriotomy), bleeding (unligated lumbar vein; **no Cell Saver**), transient LFT rise (peak 2–3 d), CPB cardiac/neuro events.
- Urine leak**: drain output >50 mL/24 h → fluid creatinine > serum. Drain → IR drain → DJ stent + Foley; most resolve 4–6 wk. Remove drain by 48 h if low output.
- Bleeding / pseudoaneurysm**: stable → bed rest, serial Hb; significant → **angiography** (better than CTA) + **selective embolisation**; refractory → re-explore.
- AVF**: bruit, haematuria, heart failure → observe / coil / stent / surgery.

7 · Theory answer frames

Q: Management of RCC with IVC tumour thrombus

- Intro** – 5–10% RCC; untreated fatal.
- Imaging** – MRV ≤ 1 mo; bland vs tumour; TEE.
- Classification** – Mayo levels + bland groups (tables §4).
- Preop** – anticoagulation, no suprarenal filter, team.
- Level-wise technique** – control order, clamp trial, Pringle, bypass.
- Reconstruction** – primary/patch/graft.
- Complications** – air/PE, bleeding, hepatic, CPB.

Q: Open partial nephrectomy — indications and technique

1. **Indications/CI** – T1, imperative; relative CI.
2. **Planning** – 3D, IOUS, hereditary.
3. **Ischaemia** – warm vs cold, adjuncts, evidence table.
4. **Excision** – enucleation vs resection.
5. **Renorrhaphy** – 2 layers.
6. **Complications** – leak, pseudoaneurysm, AVF, renal insufficiency.

Q: Incisions for open renal surgery

1. **Choice factors** – pathology, habitus, prior surgery, spine/lung.
2. **Positioning** – supine vs flank risks.
3. **Anterior vs flank incisions** – table §2.
4. **Nerve/pleura protection.**

Q: Complications of open renal surgery

1. **Vascular** – pedicle loss, SMA/SMV.
2. **Visceral** – side-specific (§6 table).
3. **Pleural** – test, repair.
4. **Procedure-specific** – chyle, thrombectomy, PN.

8 · Viva corner

Q. Why ligate the artery before the vein?

→ Kidney decompresses, collaterals fall; vein becomes flaccid, can be checked for thrombus.

Q. Right artery hidden by a large tumour?

→ Ligate interaortocavally at aortic origin, retracting LRV cephalad.

Q. Pulsatile stump after "ligating" left renal artery?

→ Could be SMA — lies 1–2 cm above LRV; or accessory artery/calcified incomplete ligation.

Q. Why not clip lumbar veins on the IVC side?

→ Clips prevent caval clamp occlusion and obstruct vision.

Q. Why spare the adrenal?

→ Mets <5%; contralateral mets equally common; routine removal risks insufficiency.

Q. Chylous ascites prevention?

→ Clip lymphatics above the renal vein (cisterna chyli).

Q. Level III: when bypass?

→ Failed clamp trial (CO ↓ >50%, MAP ↓ >30%), non-occlusive thrombus, cardiac/hepatic dysfunction.

Q. Why occlusive thrombus tolerates clamping?

→ Azygos/lumbar collaterals already developed.

Q. Why release caudal clamp first?

→ Flushes air and debris before cranial release — prevents air embolism.

Q. Separate infrarenal bland thrombus?

→ Interrupt IVC (clip/staple), don't extract; lumbar preserved as collaterals.

Q. Does clamp time matter in PN?

→ Within <30 min, preserved parenchyma dominates function.

Q. Why no Cell Saver in thrombectomy?

→ Disseminates tumour cells.

Q. Delayed haematuria after PN?

→ Pseudoaneurysm/AVF → angiography + selective embolisation.

9 · MCQ one-liners

1. CKD-EPI 2021 omits race; preferred over MDRD.
2. Stauffer syndrome: paraneoplastic hepatic dysfunction in RCC; screen with coagulation.
3. Thrombus imaging must be ≤ 1 month old.
4. Post-embolisation syndrome lasts up to 72 h.
5. SSI prophylaxis: cefazolin single dose within 1 h.
6. Subcostal flank: poor hilar access — not for PN.
7. Supracostal incision popularised by Turner-Warwick.
8. Right renal artery: posterior to renal vein and IVC.
9. SMA: 1–2 cm cephalad to LRV.
10. Left gonadal and adrenal veins drain into LRV.
11. Robson 1969 defined radical nephrectomy.
12. Adrenal metastasis $< 5\%$; imaging misses 20–25%.
13. Node positivity $\approx 5\%$; short axis ≥ 1.5 cm $\approx 50\%$ positive.
14. EORTC LND RCT: no OS benefit.
15. Venous thrombus in 5–10% RCC.
16. Level I thrombus ≤ 2 cm above renal vein ostium.
17. Level III defined by major hepatic veins, not accessory veins.
18. Level IV: supradiaphragmatic $\rightarrow$ CPB/DHCA.
19. Budd–Chiari: contraindication to thrombectomy.
20. Pringle ≤ 60 min; ideally ≤ 20 min.
21. Suprahepatic clamp: preload $\downarrow 60\%$, CO $\downarrow 40\%$.
22. DHCA core 16–18 °C.
23. Patch width = diameter $\times$ % resected $\times 3$ (Leibovich–Bower).
24. IVC graft: PTFE 16–20 mm; lifelong aspirin.
25. Bland group D (mixed) $\rightarrow$ infrarenal IVC resection.
26. Multifocality: papillary $\sim 10\%$, clear cell $\sim 2\%$.
27. Hereditary RCC: operate at 3 cm, except HLRCC.
28. Hyperfiltration injury when nephron mass reduced $> 80\%$.
29. PN relative CI: $< 20\%$ nephron mass retained.
30. Resection margin 5–10 mm; enucleation along pseudocapsule.
31. Hypothermia, mannitol, off-clamp: no benefit in RCTs.
32. Urine leak: drain creatinine $>$ serum; most resolve 4–6 wk.
33. Pseudoaneurysm: angiography preferred over CTA.
34. SMV injury: never ligate; avoid primary abdominal closure.

MCQ

142 Laparoscopic and Robotic Surgery of the Kidney

Reading time ≈ 29 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★

1 · Core concepts & patient selection

- First lap nephrectomy: **Clayman, 1991** (85-yr-old woman) **MCQ**. MIS vs open: less pain, blood loss, convalescence; better cosmesis; **equivalent long-term outcomes**. Indications = same as open.
- GA mandatory**. Pneumoperitoneum ↓ ventilation, venous return, cardiac output. High risk: COPD, intracranial pathology, cardiac disease.
- Severe COPD with hypercarbia → ↓ insufflation pressure → helium → convert to open. AirSeal (valveless, low-pressure high-flow) ↓ CO₂ use.
- Robot: bulky, arm collisions; **docking/undocking delays emergency access**; console surgeon may miss patient slipping → team communication.
- Obesity: **BMI a poor predictor**; **visceral adipose tissue** better index of difficulty. Elderly: benefits irrespective of age.
- Prior surgery alters trocar sites/access choice; former absolute contraindications now relative.

Patient must tolerate the **pneumoperitoneum**, not the renal operation → cardiopulmonary reserve, not age/BMI, governs fitness.

2 · Access & port placement

	Transperitoneal	Retroperitoneal
Space	Large; many trocar options	Limited; poor triangulation
Landmarks	Familiar	Less familiar; lens smudging
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	After colon reflection	Direct, esp. posterior
Preferred for	Most surgery; suturing; anterior tumours	Posterior PN, pyeloplasty, cyst, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity
Outcome	Reference	Shorter stay; comparable oncology

- Positioning (TP)**: modified flank, 30–45°; pad pressure points, tape, test tilt; prep whole abdomen for conversion.
- Lap TP ports**: Veress or Hasson → 3–5 trocars: 10 mm umbilical camera; 5/10 mm midline ~2 cm below xiphoid; 12 mm AAL at umbilicus (staplers). Short patient → 12 mm to lower midline; obese → shift lateral. Accessory: low midline (colon), 5 mm subxiphoid (liver).
- Robotic TP**: 3D, wristed, tremor-free, motion scaling. Arm spacing **8 cm (Si) / 6 cm (Xi)**; 12 mm low-midline assistant port (5 mm if no stapler, e.g. pyeloplasty). Linear config along lateral rectus sheath.
- Principle: ports = **broad-based triangle** — camera apex, working ports base, target far point.
- Retroperitoneal**: full flank, table flexed, axillary roll; incision posterior axillary line midway 12th rib tip–iliac crest → finger/balloon dissection (glove balloon 600–800 mL) → yellow retroperitoneal fat, smooth psoas confirm plane. Robot docked over the head.
- Balloon inflated in muscle → **flank hernia**. Too anterior → peritoneal/colon injury; too posterior → quadratus/psoas bleeding. Bail-out: open peritoneum → TP.
- HAL**: tactile feedback; popular in donor nephrectomy; scarred kidneys (XGP); **emergency finger control of bleeding**. Non-dominant hand in; forearm compression 30–100 mmHg (paraesthesia); more wound infection/incisional hernia; hand-port metastases → bag specimen.
- LESS/NOTES/SP robot**: fewer incisions buy cosmesis/pain relief at cost of triangulation. da Vinci SP (FDA 2018): 27-mm guide, 3 instruments + flexible camera; 10–25 cm cannula-to-target.

3 · Operative steps — nephrectomy

Reflect colon (Toldt) → ureter/gonadal → hilum → artery then vein → upper pole (adrenal ± spared) → ureter/lateral attachments → bag → extract

- **Left colon:** Toldt from below lower pole to above upper pole; divide lienocolic ligament. **Keep lateral attachments** (kidney otherwise flops over hilum). Avascular plane = bright-yellow mesenteric vs pale-yellow Gerota fat. Repair mesocolic defect (internal hernia).
- **Right:** liver retractor; cold sharp dissection near duodenum; **Kocher** manoeuvre → IVC.
- **Ureter:** psoas → gonadal vessels met first → ureter just posterior (peristalsis). Don't divide early — handle to lift kidney. Stay above psoas fascia (thigh numbness). Right gonadal vein tear at IVC → bleeding.
- **Hilum:** lift kidney anterolaterally → vein; artery usually posterior. **Artery first, then vein** (GIA stapler covering whole vessel). Left renal vein stapled lateral to adrenal vein. **No clips in staple line** (misfire).
- Clips ≡ staplers for safety (meta-analysis). Polymer clips: 2 mm apart, few mm from cut end. **En bloc hilar stapling:** no AV fistula at ~2 yr; saves ~45 min.
- **Radical** = simple + Gerota's intact. En bloc adrenalectomy: ligate renal vein medial to adrenal vein. Retroperitoneal left: lumbar vein usually divided to reach hilum.
- **Extraction:** intact (extended port/Pfannenstiel) or morcellation in **LapSac** (impermeable even after morcellation); change gloves/instruments; keep pneumoperitoneum and view of sac. Morcellation loses size, capsule, renal-vein and depth-of-invasion staging.
- No heavy lifting 4–6 wk if extraction incision.

	Simple	Radical
Gerota's fat	May be left	Removed en bloc
Pathology	Benign, inflammatory	RCC
Difficulty/conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple nephrectomy indications:** chronic pyelonephritis, obstructive/reflux nephropathy, TB, MCDK, renovascular HTN, ACKD, symptomatic ADPKD, post-transplant HTN. Difficulty predictors: prior pyonephrosis, nephrostomy tube; stone disease highest complications. Rescue: hand assist or **subcapsular** nephrectomy.
- **Radical results:** MIS ≡ open oncology (5/10 yr), less morbidity; stage I–II: OS advantage (database). RP vs TP RCTs: only consistent RP advantage = **shorter operative time**.
- Large tumours (>10 cm): feasible; conversion higher lap than robotic. **LND:** staging value, no OS benefit; AUA — dissect clinically positive nodes. Templates: right paracaval/precaval/retrocaval/interaortocaval; left para-/pre-aortic; crus → bifurcation.
- **IVC thrombus** (experts): early arterial ligation → milk thrombus → staple beyond or Satinsky IVC cuff. Long thrombus: Rummel tourniquets, sequential clamping infrarenal IVC → contralateral renal vein → suprarenal IVC; level III: liver mobilisation. Preop angioembolisation helps robotic (not open). → [Ch 141](#)
- Cytoreductive lap NX: quicker recovery → earlier systemic therapy.

4 · Partial nephrectomy

- PN = default where feasible. First lap PN **Winfield 1993 MCQ**. **RENAL score:** Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location vs polar lines — predicts complications, WIT, conversion.
- cT1b+/cT2: PN better function; equal oncology. **Caveat: >5 cm upstaged to pT3a recur more after PN.** Heminephrectomy = ≥30% parenchyma.
- **Central** = abutting sinus fat/collecting system; **hilar** = contact with renal artery/vein → enucleation; superficial renorrhaphy over bolster.
- Multifocal: NSS (contralateral recurrence risk). Solitary kidney: robotic PN — short WIT, rare conversion.
- TP vs RP PN: choose by **location** (anterior TP, posterior RP) + experience; RP series biased to simpler tumours.

Intraop US → open Gerota's away from lesion → score capsule → clamp → cold-scissor excision → close vessels/collecting system → renorrhaphy (sliding-clip/barbed) → unclamp

- **Trifecta:** negative margin + no major complication (30 d) + **WIT ≤25 min**. Predictors: BMI, eGFR, size, complexity.
- **Enucleation** (pseudocapsule plane): less blood/function loss; more +margins; same survival — familial, multifocal, solitary, CKD.

- Renorrhaphy: suture most reliable; sliding-clip (Lapra-Ty); barbed suture ↓ WIT; **single layer** → better function, fewer pseudoaneurysms. Sealants (Floseal, Tisseel, Coseal...): no RCT benefit.
- **Collecting system entered ~60%** (large, endophytic). Repair 2-0/3-0 Vicryl; **always drain**; routine ureteric catheter doesn't prevent fistula.
- Historical WIT limit **30 min**; **preserved parenchymal volume predicts function better than WIT**.

Hilar strategy	Best for	Verdict
Artery-only vs artery+vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective/super-selective (NIRF, duplex)	Mapped segmental supply	RCT: no 6-mo benefit (futility)
Off-clamp (zero ischaemia)	Small, exophytic	Early eGFR better; RCTs no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia (ice slush, cold perfusion)	Solitary kidney, long WIT	Benefit unproven

- Robotic vs lap PN: fewer conversions, shorter WIT. Robotic vs open: less blood loss, fewer complications.
- **+Margins 0-7%**: raise recurrence only in high-risk (pT2-3, grade 3-4) → surveillance, not reoperation.
- **Port-site recurrence ~0.1%**; not due to pneumoperitoneum/gas; prevent with impermeable bag, no spillage.
- Local recurrence after RN ~2%: resection + systemic. Salvage after PN/ablation: planes lost (no pseudocapsule/fibrosis) → often RN.

5 · Other procedures

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, ? malignancy	Aspirate first (proves pain source); excise wall, biopsy, fulgurate base; collecting system entered → close + drain. ADPKD: decortication, bilateral NX, renal denervation
Lap renal biopsy	Failed percutaneous, bleeding diathesis, solitary kidney, morbid obesity	Retroperitoneal, 2 ports; more glomeruli
Nephropexy	Ptois >5 cm/2 vertebrae supine→erect + pain, ↓ Doppler flow or obstruction	Right, slim young women; 2-0 Prolene capsule to psoas/quadratus fascia; avoid iliohypogastric/genitofemoral n.
Calyceal diverticulectomy	Endoscopy failed	Neck sutured closed , cavity obliterated + fat (endoscopy widens neck)
Nephrolysis	Chyluria, loin pain-haematuria	Lateralise on cystoscopy; hilar lymphatic ligation ± omental wrap
Ablation	≤3 cm, comorbid, solitary, hereditary	Percutaneous now preferred. Cryo: 2 freeze-thaw, iceball 0.5-1 cm beyond, real-time US. RFA: 60-100°C (<105°C), 1 cm margin, impedance-guided; more retreatment

6 · Complications

- Overall **~15%**; PN > RN > simple. Learning curve ~20 lap nephrectomies. Robotic PN: half the open conversion of lap. Age not a major risk; conversions cluster in infection.
- **Vascular injury = commonest**; major <1/1000, fatal 8-17%; at **access (Veress/first trocar)** or hilar dissection. Device failure: staplers > non-locking Ti clips > locking clips; stapler errors mostly operator (over clips, thick tissue, early stop).
- **SMA mistaken for left renal artery** → bowel necrosis (obese, large medial tumours).
- End of case: **lower insufflation pressure** to unmask venous bleeding.
- Haematuria after PN (~day 12) → **pseudoaneurysm/AVF** → selective embolisation.
- Urine leak 0.8-10%: drain → stent/nephrostomy.

- **Unrecognised bowel injury** (<1%, mortality ~3%): thermal ~half, 32% at access; **pain at one trocar site**, distension, low-grade fever, **normal/low WBC**, bowel sounds present → CT oral contrast. Thermal presents later.
- Diaphragm injury: ↑ airway pressure → purse-string + suction tube on inspiration. Unrecognised pneumothorax ~0.3%; no routine CXR.
- Spleen/pancreas (left), liver/gallbladder (right). Rhabdomyolysis (full flank, long cases) → CK, hydration. Port-site/incisional hernia, ileus, DVT (risk-stratified prophylaxis).
- Prior surgery not a contraindication → Veress away from scars, Hasson or retroperitoneal.

Major bleed: locate → temporary control → inform anaesthetist (blood, open set) → ↑ pressure/add trocar → clip/suture (never blind) → else convert + vascular surgeon

7 · Theory answer frames

Q: Transperitoneal vs retroperitoneal laparoscopic renal surgery

1. **Access & positioning** – modified vs full flank; Veress/Hasson vs balloon.
2. **Ports** – triangle principle; habitus adjustments.
3. **Comparison** – table §2.
4. **Evidence** – RP shorter operation, equal oncology.
5. **Selection** – posterior tumour, adhesions, dialysis → RP.

Q: Steps of laparoscopic radical nephrectomy

1. **Preop** – imaging maps hilum; fitness for pneumoperitoneum.
2. **Access/ports** – §2.
3. **Steps** – algo §3; artery before vein; adrenal handling.
4. **Extraction** – bag, intact vs morcellation.
5. **Special** – large tumours, LND, IVC thrombus.
6. **Complications** – §6.

Q: Robotic partial nephrectomy — hilar control and outcomes

1. **Selection** – RENAL score; central/hilar.
2. **Steps** – algo §4.
3. **Ischaemia** – table; parenchyma volume > WIT.
4. **Renorrhaphy & collecting system**.
5. **Outcomes** – trifecta; margins; pseudoaneurysm, leak.

Q: Complications of laparoscopic renal surgery

1. **Incidence/risk factors** – 15%, learning curve.
2. **Vascular** – access, hilar, stapler; algo.
3. **Bowel** – presentation, CT, management.
4. **Solid organ, diaphragm, urinary, neuromuscular, hernia**.

8 · Viva corner

Q. Why keep lateral attachments early?

→ Otherwise kidney rolls medially and obscures the hilum.

Q. Why not divide ureter early?

→ It is the handle to lift the kidney and put hilum on stretch.

Q. Why artery before vein?

→ Standard sequence; early arterial ligation reduces collateral flow and may retract venous thrombus.

Q. Clip in stapler path?

→ Stapler misfire → major bleeding.

Q. Why is simple nephrectomy harder than radical?

→ Inflammation obliterates planes; higher conversion; subcapsular rescue.

Q. Retroperitoneal access indications?

→ Posterior tumours, dense adhesions, peritoneal dialysis, morbid obesity, pregnancy.

Q. Does WIT matter most?

→ No — preserved parenchymal volume predicts function better.

Q. Positive margin after PN?

→ Surveillance; recurrence rises only in high-risk tumours.

Q. Bowel injury presentation?

→ Trocar-site pain, low-grade fever, low/normal WBC → CT with oral contrast.

Q. Haematuria day 12 post PN?

→ Pseudoaneurysm → angioembolisation.

Q. COPD patient becomes hypercarbic?

→ Lower pressure → helium → convert.

Q. Role of hand port in crisis?

→ Finger compression while conversion is prepared.

Q. Nephroptosis — when to operate?

→ Only for proven pain, ischaemia or obstruction, not descent alone.

9 · MCQ one-liners

- | | |
|---|---|
| 1. First lap nephrectomy: Clayman 1991; first lap PN: Winfield 1993. | 14. >5 cm tumour upstaged pT3a: more recurrence after PN than RN. |
| 2. Better predictor of difficulty than BMI: visceral adipose tissue. | 15. Cryo iceball 0.5–1 cm beyond tumour; two freeze-thaw cycles. |
| 3. Robot arm spacing: 8 cm Si, 6 cm Xi. | 16. RFA 60–100°C; >105°C vaporises tissue. |
| 4. Retroperitoneal RN consistent advantage: shorter operative time. | 17. Commonest MIS complication: vascular injury. |
| 5. Gonadal vessels lie anterior to ureter; right gonadal vein tears at IVC. | 18. Device failure: staplers > non-locking clips > locking clips. |
| 6. Renal artery usually posterior to vein. | 19. Bowel injury mortality ~3%; thermal causes ~half. |
| 7. En bloc hilar stapling: no AV fistula; saves ~45 min. | 20. Unrecognised pneumothorax ~0.3%. |
| 8. Trifecta WIT ≤25 min; historical WIT limit 30 min. | 21. Neuromuscular injury 2.7% of lap urological surgery. |
| 9. Collecting system entered in ~60% lap PN. | 22. Learning curve ~20 lap nephrectomies. |
| 10. Single-layer renorrhaphy: fewer pseudoaneurysms. | 23. Robotic PN volume plateau 18–20 cases/yr. |
| 1. Positive margins 0–7% of MIS PN. | 24. Nephroptosis: >5 cm or 2 vertebral bodies; asymptomatic in 20% women. |
| 2. Port-site recurrence ~0.1%. | 25. LapSac impermeable to tumour cells even after morcellation. |
| 3. Local recurrence after RN ~2%. | 26. Hand-port forearm pressure 30–100 mmHg. |

143 Focal Therapy for Renal Tumours

Reading time ≈ 20 min (slow pace) · Theory ★★ · Viva ★★★ · MCQ ★★★

1 · Indications & guidelines

- Rationale: small RCC seldom kills; CKD and comorbidity do. **PN = gold standard** for cT1a. SRM biology/biopsy → [Ch 137](#)
- **Indications:** high surgical risk/comorbid, solitary kidney, bilateral tumours, hereditary (VHL: repeated treatments expected).
- Advantages: no dissection/renorrhaphy → faster recovery, fewer complications; renal function ≈ or better than PN.
- **Biopsy before ablation** (AUA 2021: universal; separate prior session preferred): fewer non-diagnostic results, fewer benign ablations, informed consent. Routine post-ablation biopsy – no value.

Counsel (age, comorbidity, CKD risk, size/location) → biopsy → re-counsel (histology, staging) → PN / surveillance / percutaneous ablation (only after biopsy discussed)

Guideline	Size ceiling / position
AUA 2021	cT1a ≤3 cm , esp. comorbid; alternative (PN prioritised); percutaneous preferred
EAU 2022	TA not routinely >3 cm ; CA not routinely >4 cm ; frail/comorbid
NCCN 2022	Only body allowing cT1b (non-surgical candidates)
ASCO 2017	Option if complete ablation achievable

2 · Mechanisms

- All energies: **central lethal zone** (direct injury/necrosis) + **peripheral sublethal zone** (apoptosis, ischaemia, immune response).
- **Cryo** cell death: freeze (extracellular ice → osmotic dehydration → intracellular ice) + thaw (fluid shift → oedema, lysis) + vascular thrombosis + apoptosis + immunomodulation (DAMPs → dendritic cells → T cells).
- Cryo lethal: normal parenchyma -19.4°C ; tumour up to -50°C → target **$\leq -40^{\circ}\text{C}$** . Iceball edge 0°C ; $<-20^{\circ}\text{C}$ only 3 mm inside edge → iceball **5-10 mm beyond tumour**.
- Heat: 40-45 $^{\circ}\text{C}$ sublethal; **$\geq 60^{\circ}\text{C}$** protein denaturation/coagulative necrosis; $\geq 100^{\circ}\text{C}$ vaporisation, charring. RFA lethal: 45 $^{\circ}\text{C} \times 60$ min, 55 $^{\circ}\text{C} \times 5$ min, **70 $^{\circ}\text{C} \times 1$ min**.

	Cryoablation	RFA	MWA
Physics	Argon expansion (Joule-Thomson); helium for active thaw	AC 450-1200 kHz → ionic friction (heat made in tissue)	915 MHz / 2.45 GHz → dielectric hysteresis (water dipoles)
Target	$\leq -40^{\circ}\text{C}$	$\geq 70^{\circ}\text{C}$ uniformly; generator 90-105 $^{\circ}\text{C}$ (≤ 105)	Often $>100^{\circ}\text{C}$
Cycles	Double freeze-thaw , 8-10 min (e.g. 10 + 6-8)	Two, ≥ 30 -s cool-down	Single time-power (e.g. 65 W $\times$ 5 min)
Margin	Iceball 5-10 mm	≥ 5 mm	≥ 5 mm
Charring limits	No	Yes (impedance rise)	No
Grounding pad	No	Yes (monopolar)	No
Heat sink	Yes	Yes	Least
Real-time zone	Iceball visible (CT/MRI/US)	Not visible – impedance/temperature only	Gas bubbles on US

	Cryoablation	RFA	MWA
Pros	Visible zone; spares collecting system → central tumours	Long follow-up data	Fast, 2 cm active heating, multiple antennas, lowest complications (NMA)
Cons	Bleeding (no coagulation), tumour fracture	Charring, heat sink, grounding-pad burns	Speed endangers adjacent organs; shortest follow-up

- RFA generators: temperature- vs impedance-controlled – neither superior. Multitine electrodes clinically best. Wet RFA (saline) → larger but less controlled zone; bipolar → elliptical zone, not adopted.
- Probe cooling: RFA – prevents charring; MWA – protects track, allows higher power.

3 · Technique

- **Percutaneous = standard** (laparoscopic declined since 2010). Lap vs perc: same efficacy; perc less morbidity, stay, cost. Laparoscopic only for prior scarring or failed hydrodisplacement.
- GA preferred (controls respiration; high-frequency jet ventilation ~120/min reduces renal motion). IV antibiotics → prone/flank on CT → US/CT placement → non-contrast CT check → ablate → contrast CT.
- **Hydrodisplacement**: spinal needle between tumour and bowel/ureter/organ → saline (**D5W for RFA** – non-conductive) ± contrast; aim e.g. colon >1 cm away. Caution in poor cardiac function.
- **Pyeloperfusion** (central/near ureter): ureteral catheter with saline **opposite** to ablation – cold for RFA/MWA, warm for CA.
- CA: multiple probes **1-2 cm apart**; remove only after thaw, without resistance. MWA antennas placed tangentially to collecting system; ≤2.5 cm → 1 antenna, 2.5-3.5 → 2, 3.5-4.5 → 3.

4 · Follow-up, success & salvage

- **Success = complete loss of contrast enhancement** on CT/MRI.
- **Incomplete ablation**: persistent enhancement on *initial* imaging → repeat ablation.
- **Local recurrence**: initial non-enhancement, *later* enlargement and/or enhancement → biopsy ± surveillance, re-ablation or extirpation.
- **Enlargement, any modality = recurrence until proved otherwise.**
- Appearance: CA shrinks >50% in 1st year; RFA minimal contraction, **fibrotic halo**; MWA low-attenuation, contracts; MRI T1 hypointense, non-enhancing.
- AUA schedule: CT/MRI ± contrast + chest imaging at **6, 12, 24, 36, 48, 60 mo**, then shared decision. US not routine (unless CEUS).

5 · Oncological outcomes vs PN

- No RCT of TA vs surgery. LRFS worse after TA but **equal once re-ablation counted; MFS and CSS no different**; OS lower (older, comorbid cohort).
- Size drives failure: RFA worse ≥3 cm; T1b worse than T1a.
- No modality oncologically superior; choice by institutional preference/tumour.
- Biases favouring TA: benign tumours included, under-graded biopsies. Against TA: occult pT3a, growth-selected tumours.
- CSS: CA 80-100% at 10 yr; RFA ≈98%. OS 75-85% at 5 yr (comorbidity, not cancer).
- EORTC 30904 (PN vs RN, ≤5 cm): only 12 RCC deaths at 9.3 yr → small-RCC trials need huge cohorts.

Ablation replaces the pathological margin with an **imaging margin** – hence biopsy first and structured, prolonged surveillance.

6 · Complications

- Major (Clavien ≥3) **≤5%**; fewer than PN; perc < lap. Predictors: RENAL score, ASA >3 (~3×), (MC)² score (CA only).

Complication	Key points
Haemorrhage - most common	Transfusion ~2-3%; multiple probes, CA. Full thaw before probe removal. Persistent → selective embolisation
Collecting system/ureter	Central tumours; urinoma, fistula, UPJ stricture. Prevent: stent + pyeloperfusion, prefer CA. Treat: stent, drain
Adjacent organ/bowel	Tumour <1 cm from anterior organ. Hydrodisplacement; colonephric fistula → stent + low-residue diet; peritonitis → explore
Pleural	Probe above 12th rib → pneumothorax; aspirate; chest tube rarely
Nerve/pain	Posterior tumours → neuralgia; grounding-pad burns (RFA) – pads at same level on posterior thighs
Infection	<1% but may be lethal; ileal conduit/colonised urine; abscess at 1-6 mo. Pre-op urine culture

7 · Newer modalities (all investigational)

Modality	Mechanism	Limitation
HIFU	Focused US; thermal + cavitation; fully extracorporeal	Incomplete ablation (respiration, acoustic interference)
Histotripsy	Focused US, mechanical, non-thermal	No clinical data (HOPE4KIDNEY ongoing)
SBRT/SABR	High dose, 1-few fractions, motion tracking; no GA, no stopping anticoagulants	No imaging endpoint; viable attenuated tumour on biopsy; lower CSS
IRE	Electrical pulses → nanopores; saves matrix, ducts, vessels	Suboptimal oncology

8 · Theory answer frames

Q: Compare cryoablation, RFA and MWA for small renal masses.

1. **Indications** – §1 table.
2. **Mechanism & parameters** – comparison table §2.
3. **Technique & adjuncts** – perc, hydrodisplacement, pyeloperfusion.
4. **Outcomes** – equivalent oncology; MWA fewest complications.

Q: Follow-up and management of recurrence after renal ablation.

1. **Success definition** – loss of enhancement.
2. **Incomplete vs recurrence** – timing of enhancement.
3. **Imaging by modality; AUA schedule.**
4. **Salvage** – re-ablation, PN/RN, SABR in comorbid.

Q: Complications of percutaneous renal ablation.

1. **Incidence & predictors.**
2. **Specific complications** – table §6.
3. **Prevention** – planning, displacement, pyeloperfusion.

9 · Viva corner

Q. Why a 5-10 mm iceball margin?

→ Edge is 0 °C; lethal –40 °C lies well inside the visible iceball.

Q. Why double freeze-thaw?

→ Enlarges necrotic zone; thaw itself lethal.

Q. Why D5W, not saline, for RFA hydrodissection?

→ Saline conducts current; D5W is non-ionic.

Q. Why does MWA beat heat sink?

→ Faster heating, radiates through desiccated tissue/blood, active heating to 2 cm.

Q. Central tumour near collecting system - modality?

→ CA (spares collecting system) + stent with warm pyeloperfusion.

Q. How is RFA monitored?

→ Impedance/temperature; zone not visible. Quadrant $\geq 70^\circ\text{C}$ during cool-down excludes heat sink.

Q. Enhancement at 6 months after initial non-enhancing CT?

→ Local recurrence → biopsy; re-ablate or extirpate.

Q. Why is TA-PN comparison biased?

→ Older comorbid patients, benign/under-graded tumours, occult pT3a, long natural history.

Q. When laparoscopic ablation?

→ Prior scarring or failed hydrodisplacement.

Q. Why is SBRT experimental?

→ No imaging endpoint, unclear fractionation, viable tumour on biopsy.

10 · MCQ one-liners

1. AUA: consider TA for cT1a ≤ 3 cm.
2. Only NCCN allows TA for cT1b.
3. EAU: CA not routinely >4 cm.
4. Argon cools, helium warms – Joule-Thomson.
5. Cryo target $\leq -40^\circ\text{C}$; normal kidney dies at -19.4°C .
6. $< -20^\circ\text{C}$ only 3 mm inside iceball edge.
7. RFA frequency 450–1200 kHz.
8. MWA frequencies 915 MHz and 2.45 GHz.
9. MWA mechanism: dielectric hysteresis.
10. $>70^\circ\text{C}$ for 1 min lethal to RCC.
11. RFA temperature kept $\leq 105^\circ\text{C}$ to avoid charring.
12. Only RFA needs grounding pads.
13. MWA least susceptible to heat sink.
14. MWA zone size $\propto$ ablation time.
15. RFA post-ablation: fibrotic halo, minimal shrinkage.
16. CA lesion shrinks $>50\%$ in first year.
17. Pyeloperfusion fluid temperature opposes ablation.
18. Most common complication: haemorrhage.
19. Major complications $\leq 5\%$.
20. MWA lowest complication rate (network meta-analysis).
21. CSS/MFS equal TA vs PN; LRFS equal after re-ablation.
22. Histotripsy: non-thermal; HIFU: thermal + cavitation.
23. IRE preserves extracellular matrix and vessels.
24. Outpatient ablation: ≤ 3 cm low complexity, no anticoagulation, few comorbidities.
25. HIFU threshold for cavitation $>3500\text{ W/cm}^2$.

144 Treatment of Advanced Renal Cell Carcinoma

Reading time ≈ 34 min (slow pace) · Theory ★★★ · Viva ★★★ · MCQ ★★★

1 · Core concepts & prognostic models

- ~1/3 present with synchronous mets; further **20-40%** of localised tumours later metastasise. mRCC 5-yr survival ≈ **12%**; 10-yr <5%.
- Three eras: cytokine → single-agent targeted → **immune checkpoint (IO) combinations**.
- Better outcome: long diagnosis-to-metastasis interval, few sites. Worse: poor PS, nodal/liver mets.
- Pathology, staging, nomograms for localised disease → [Ch 137](#).

Factor	MSKCC / Motzer (5) - cytokine era, n=670	IMDC / Heng (6) - VEGF-TKI era, n=849
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< LLN	< LLN
Corrected calcium	>10 mg/dL	> ULN
LDH	>1.5 × ULN	—
Time factor	No prior nephrectomy	<1 yr diagnosis → systemic Rx
Neutrophils / platelets	—	> ULN (each)
Favourable (0) median OS	20 mo	43.2 mo
Intermediate (1-2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

3 shared factors (PS, Hb, Ca). IMDC drops LDH & nephrectomy, adds neutrophilia, thrombocytosis, <1 yr interval. Poor risk = ≥3 in both. IMDC risk group now **selects first-line regimen**.

2 · Biology & drug targets

- **VHL** (3p) tumour suppressor – two-hit. pVHL + elongin B/C + CUL2 → ubiquitinates **HIF-α** (after prolyl hydroxylation in normoxia).
- VHL loss (mutation or promoter hypermethylation – most sporadic ccRCC) → HIF accumulates even in normoxia → VEGF, PDGF, TGF-α, GLUT-1, EPO.
- mTOR regulates HIF-1α translation. MET/AXL (HIF-driven) and FGFR mediate resistance to VEGF blockade → broader TKIs.
- Checkpoints: CTLA-4 (CD28 homologue; binds B7) and PD-1 (T cell) / PD-L1, PD-L2 (tumour, macrophages) inhibit T-cell responses. VEGF inhibitors are immunomodulatory (sunitinib ↓ Tregs; ↓ suppressive macrophages) → rationale for IO/TKI.
- ccRCC is chemoresistant (response 5.5-6%; MDR gene proposed, MDR inhibitors failed) and hormone-insensitive (medroxyprogesterone ~2%).

Agent	Class / target	Signature toxicity / note
Bevacizumab	mAb vs VEGF-A ligand	Bleeding, proteinuria, HTN
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, HFS; now rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; 50 mg, 4 on/2 off	Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, hypothyroidism
Pazopanib	VEGFR, PDGFR, KIT	Hepatotoxicity (↑ALT) ; less fatigue/HFS
Axitinib	VEGFR1-3 only (most selective)	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1-3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, MET, AXL	Fatigue, diarrhoea, PPE; frequent dose reduction

Agent	Class / target	Signature toxicity / note
Lenvatinib	VEGFR1-3, FGFR1-4 , PDGFR- α , RET, KIT	Paired with everolimus or pembrolizumab
Temsirolimus	mTOR, IV weekly	Mucositis, rash, $\uparrow$ glucose, $\downarrow$ PO ₄ , $\uparrow$ cholesterol, pneumonitis
Everolimus	mTOR, oral daily	Asthenia, anaemia
Belzutifan	HIF-2α inhibitor , oral	Post-IO + post-TKI; VHL-associated RCC
Nivolumab / pembrolizumab	Anti-PD-1	Autoimmune: diarrhoea/enteritis, hypophysitis, vitiligo
Avelumab	Anti-PD-L1	
Ipilimumab	Anti-CTLA-4	

3 · Surgery in metastatic disease

Cytoreductive nephrectomy (CN)

- Rationale: rare spontaneous regression of mets (<1%, often transient); bulky primary suppresses T cells; cytokines rarely shrink primary. CN alone does not alter outcome.
- Cytokine era: **SWOG 8949 & EORTC** – CN + IFN- α improved OS **without changing response rate** (combined 13.6 vs 7.8 mo).
- TKI era: **CARMENA** – sunitinib alone noninferior to CN → sunitinib in MSKCC intermediate/poor risk. **SURTIME** – deferred CN after sunitinib safe, OS favoured deferral.
- CARMENA caveats: ~45% MSKCC poor risk; ~25% of CN arm did not get intended Rx; 17% of sunitinib arm had later CN; per-protocol noninferiority lost; low-burden patients excludable; stopped early, underpowered; sunitinib no longer standard.
- Poor candidates:** poor PS, comorbidity, rapidly progressive disease, brain/liver mets, poor risk. Authors offer CN to **well-selected** ccRCC patients (fit, low burden, favourable).

mRCC → IMDC risk + PS + burden → fit, low burden, favourable → consider upfront CN → systemic Rx
Intermediate/poor or high burden → systemic Rx first → deferred CN in responders

Metastasectomy

- No RCT; retrospective. Isolated **pulmonary** mets most often resected for cure; 5-yr survival **35-50%** after complete resection.
- Favourable: **complete resection**, solitary lesion, age <60, DFI >1 yr, lung site, size <4 cm, metachronous. All are markers of indolent biology (selection bias).

Palliative surgery

- Palliative CN: intractable pain, haematuria, constitutional symptoms, paraneoplastic (hypercalcaemia, erythrocytosis, thrombocytosis, HTN). Alternatives often suffice: medical Rx for pain/Ca; **angioembolisation** for haematuria. Only ~half of hypercalcaemic patients improve after resection.
- Palliative metastasectomy: solitary brain met, weight-bearing bones/joints, vertebral mets with impending cord/root compromise – usually combined with **radiation** $\pm$ systemic Rx.

4 · Systemic management

Cytokines

	IFN- α	High-dose IL-2
Response	10-15%	15-20%
Durable CR	<2%	7-9% ; >60% of CRs durable
Regimen	Schedule/route immaterial	600,000-720,000 IU/kg IV bolus q8h, max 15 doses; low-dose inferior (2 RCTs)

	IFN- α	High-dose IL-2
Toxicity	Constitutional	Vascular leak syndrome ; early mortality 2–5% ($\downarrow$ with fluids, pressors, antibiotics)
Role	Not used first-line	Selected fit patients (only agent giving durable CR); approved 1992 on phase II ORR

- IL-2 combos (IFN, 5-FU, TIL/LAK) add toxicity, not survival. CAIX predictive retrospectively, not prospectively.

First-line ccRCC by IMDC risk (NCCN / EAU)

Risk	Preferred / standard	Alternatives	Certain circumstances
Favourable	Pembro + axitinib; nivo + cabozantinib; pembro + lenvatinib	Avelumab + axitinib, cabozantinib, ipi + nivo, pazopanib, sunitinib (EAU if IO impossible: sunitinib, pazopanib)	Active surveillance , axitinib, HD IL-2
Intermediate / poor	Same three IO/TKI + ipi + nivo	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, HD IL-2, temsirolimus

Ipi + nivo is **not** preferred for favourable risk (sunitinib did better there in CheckMate 214). Avelumab + axitinib: FDA-approved, not NCCN-preferred (no OS benefit). PD-L1 status selects for none. IO/TKI vs IO/IO never compared head-to-head: IO/TKI $\rightarrow$ higher ORR/PFS, better tolerated; IO/IO $\rightarrow$ deeper durable responses, more autoimmune toxicity.

Subsequent lines - "switch mechanism"

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	Nivolumab or cabozantinib (alt: axitinib)	IO combos, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, belzutifan , cabozantinib, lenvatinib + everolimus, tivozanib

- IO re-challenge after IO only "useful in certain circumstances". Tivozanib: after ≥ 2 prior therapies. Belzutifan: after both PD-1/PD-L1 and VEGF TKI.
- Everolimus alone supplanted by nivolumab/cabozantinib; still used with lenvatinib.
- Failed combos: sunitinib + mTOR (toxic); nivo + sunitinib/pazopanib (toxic); bevacizumab + mTOR (RECORD-2, INTORACT – no gain).

Toxicity themes

- IO**: immune-related (enteritis, hypophysitis, vitiligo). Ipi + nivo: grade 3–4 **46%**, high-dose steroids needed 35%, permanent discontinuation 22%. Ipi 1 mg/kg + nivo 3 mg/kg $\times 4$ chosen (CheckMate 016) – higher ipi dose more toxic. Nivolumab monotherapy better tolerated than everolimus (G3–4 19% vs 37%).
- TKI**: HTN, diarrhoea, HFS/PPE, fatigue; managed by dose reduction/interruption (cabozantinib – frequent reductions, few stop). Pazopanib – caution if hepatotoxicity risk.
- IL-2**: vascular leak $\rightarrow$ hypotension, third-spacing, respiratory/multiorgan failure.

Non-clear cell RCC (15–25%)

- No agent of proven efficacy; ccRCC drugs give modest activity; **enrol in trials**. Chemo possible in sarcomatoid (gemcitabine-based, small series).
- Papillary**: sunitinib response 5–10%; erlotinib modest. Temsirolimus benefit in non-clear subgroup ($\sim 20\%$) of poor-risk trial. Everolimus (RAPTOR) modest; sunitinib vs everolimus – neither clearly better, trend to sunitinib. Chromophobe: trend to longer PFS on everolimus.
- MET**: HPRC germline activating MET (chr 7) $\rightarrow$ type 1 papillary; sporadic: somatic MET $\sim 13\%$, **trisomy 7 in $>2/3$** . Foretinib (MET+VEGFR2): $\sim$ half of germline MET carriers responded. Savolitinib, crizotinib active in MET-driven.
- HLRCC**: FH loss $\rightarrow$ fumarate $\rightarrow$ VHL-independent HIF; aggressive type 2 papillary; glycolysis-dependent $\rightarrow$ **bevacizumab + erlotinib** (glucose restriction), PR $\sim 2/3$.

- **Collecting duct:** urothelial-like → **gemcitabine + carboplatin.**

Syndrome	Gene	Tumour	Therapeutic lead
VHL	VHL (3p), LOF	Bilateral multifocal clear cell	VEGF/VEGFR, mTOR, belzutifan
HPRC	MET (7), activating	Papillary type 1	MET inhibitors
HLRCC	FH, LOF	Papillary type 2	Bev + erlotinib

5 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS ↑ (11.1 vs 8.1; 17 vs 7 mo); ORR same	Basis of CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority, n=450	OS 13.9 vs 18.4 mo, HR 0.89	No upfront CN in intermediate/poor risk
SURTIME	Immediate vs deferred CN (sunitinib first)	OS 15.0 vs 32.4 mo, HR 0.57	Systemic first, deferred CN safe
Sunitinib vs IFN	Phase III, 1st line	PFS 11 vs 5 mo; ORR 31 vs 6%	Sunitinib = comparator of all later trials
COMPARZ	Pazopanib vs sunitinib, 1st line noninferiority	PFS HR 1.05; OS same	Equivalent; differ in toxicity; QoL favoured pazopanib (PISCES preference)
CALGB 90206 / AVOREN	Bev + IFN vs IFN	PFS ↑, no OS gain	Bev not widely used alone
TARGET	Sorafenib vs placebo post-cytokine	PFS 5.5 vs 2.8; OS NS	Historical
Temsirolimus phase III	Tem vs IFN vs combo; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR-i with OS gain; combo no gain
Everolimus vs placebo	Post-VEGF TKI	PFS 4.0 vs 1.9; OS NS	Now supplanted
AXIS	Axitinib vs sorafenib, 2nd line	PFS 6.7 vs 4.7; OS NS	2nd-line axitinib
METEOR	Cabozantinib vs everolimus, post-VEGF	PFS HR 0.58; OS HR 0.70	Only TKI with OS gain over active comparator
CABOSUN	Phase II cabo vs sunitinib, IMDC int/poor	PFS 8.2 vs 5.6, HR 0.66	1st-line cabozantinib
Lenvatinib ± everolimus	Phase II, post-VEGF	Combo PFS ↑ vs everolimus	Approval of len + eve
TIVO-3	Tivozanib vs sorafenib, 3rd/4th line	PFS HR 0.73; OS NS	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus post-IO + TKI	PFS HR 0.75 (medians equal – late separation)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus post-TKI; primary OS	OS 25.0 vs 19.6, HR 0.73 ; PFS same	IO benefit shows as OS, not PFS
CheckMate 214	Ipi + nivo vs sunitinib; IMDC int/poor	OS HR 0.63; CR 9 vs 1%; PFS NS initially	Only IO/IO; not for favourable risk
KEYNOTE-426	Pembro + axitinib vs sunitinib	PFS + OS ↑; ORR 60 vs 40%	All risk groups; PD-L1 independent
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS ↑; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivo + cabo vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups

Trial	Design	Result	Take-home
CLEAR	Pembro + lenvatinib vs sunitinib (± len + eve arm)	PFS 23.9 vs 9.2, HR 0.39 ; ORR 71%	Largest PFS/ORR effect

6 · Theory answer frames

Q: Role of cytoreductive nephrectomy in metastatic RCC.

1. **Rationale** – immune suppression by primary, spontaneous regression, palliation.
2. **Cytokine era** – SWOG 8949, EORTC: OS ↑, ORR unchanged.
3. **TKI era** – CARMENA, SURTIME: results + caveats (see §3, §5).
4. **Patient selection** – good PS, low burden, favourable/intermediate IMDC, no brain/liver mets, resectable; avoid poor risk.
5. **Sequencing** – systemic first in intermediate/poor; deferred CN in responders.
6. **Palliative CN** – indications; alternatives (embolisation).

Q: Prognostic models in metastatic RCC.

1. **Why** – stratify trials; select first-line therapy.
2. **MSKCC** – 5 factors, risk groups, OS (table §1).
3. **IMDC** – 6 factors; differences from MSKCC.
4. **Clinical use** – ipi + nivo only int/poor; temsirolimus poor; surveillance favourable.

Q: Current systemic therapy of metastatic ccRCC.

1. **Biology** – VHL-HIF-VEGF; checkpoints.
2. **Drug classes** – cytokines, VEGF mAb, TKIs, mTOR-i, HIF-2α-i, IO (table §2).
3. **First line by IMDC risk** – IO/TKI vs IO/IO; key trials.
4. **Subsequent** – switch mechanism; post-IO vs post-TKI.
5. **Toxicity** – irAEs, TKI class effects, IL-2 vascular leak.
6. **Adjuncts** – CN, metastasectomy, palliative RT/surgery.

Q: Short note – non-clear cell RCC treatment.

1. **Scale** – 15–25%; no proven agent; trials.
2. **Papillary** – VEGF/mTOR modest; MET biology; MET inhibitors.
3. **HLRCC** – FH, bev + erlotinib.
4. **Chromophobe / collecting duct / sarcomatoid** – everolimus; gem-carbo; gem-based chemo.

7 · Viva corner

Q. Why did CN improve OS in cytokine trials without changing response rate?

→ Benefit not mediated through better cytokine response; likely removal of immunosuppressive bulk/primary-driven progression.

Q. Does CARMENA mean CN is dead?

→ No – mostly poor-risk, underpowered, high crossover; fit low-burden favourable patients were largely absent; sunitinib no longer standard.

Q. Which IMDC factors differ from MSKCC?

→ IMDC adds neutrophilia, thrombocytosis, <1 yr diagnosis-to-treatment; drops LDH >1.5×ULN and no prior nephrectomy; Ca cut-off ULN not 10 mg/dL.

Q. Why is IL-2 still mentioned?

→ Only agent giving durable CR (7–9%) with long-term disease-free status in >60% of CRs.

Q. Why does belzutifan work?

→ HIF-2α sits immediately downstream of VHL loss; drug binds a found pocket in a factor once "undruggable".

Q. Why does cabozantinib work after VEGF failure?

→ Also blocks MET and AXL, HIF-driven escape pathways.

Q. Why was the median PFS equal in LITESPARK-005 yet HR 0.75?

→ Non-proportional hazards – curves separate late.

Q. Which first-line regimens are restricted to intermediate/poor risk?

→ Ipi + nivo, cabozantinib, temsirolimus.

Q. Is PD-L1 needed before IO?

→ No – benefits independent of PD-L1 in KEYNOTE-426, 9ER, CLEAR.

Q. Who benefits from metastasectomy?

→ Complete resection, solitary, lung, DFI >1 yr, metachronous, <60 y, small – i.e. indolent biology.

Q. Temsirolimus vs everolimus?

→ Tem IV weekly, 1st line poor risk, OS gain vs IFN; eve oral daily, post-TKI, PFS only.

Q. Haematuria in unfit mRCC patient?

→ Angioembolisation rather than palliative nephrectomy.

Q. Why IO + VEGF-TKI synergy?

→ VEGF blockade reduces Tregs and suppressive macrophages – immunomodulatory.

8 • MCQ one-liners

1. MSKCC LDH cut-off: >1.5 × ULN; calcium >10 mg/dL.
2. IMDC poor-risk median OS 7.8 mo; favourable 43.2 mo
3. MSKCC poor-risk median OS 4 mo.
4. VHL gene: chromosome 3p; complex with elongin B/C and CUL2.
5. HPRC: MET on chromosome 7; type 1 papillary.
6. Sporadic papillary RCC: trisomy 7 in >2/3.
7. HLRCC: fumarate hydratase; type 2 papillary.
8. Most selective VEGFR TKI: axitinib (VEGFR1–3).
9. Only anti-PD-L1 in RCC combos: avelumab.
10. Lenvatinib uniquely hits FGFR1–4.
11. Sunitinib schedule: 50 mg, 4 weeks on, 2 off.
12. Sunitinib causes hypothyroidism; pazopanib raises ALT.
13. HD IL-2: 600,000–720,000 IU/kg q8h, max 15 doses.
14. IL-2 dose-limiting toxicity: vascular leak syndrome.
15. IL-2 FDA approval 1992 – only approval based on phase II response.
16. Temsirolimus: only mTOR inhibitor with OS benefit (poor risk, all histologies).
17. CheckMate 025 primary endpoint: OS (HR 0.73); PFS not improved.
18. CheckMate 214: CR 9% vs 1%; favourable risk did better with sunitinib.
19. CLEAR: PFS HR 0.39 – largest effect among IO combos.
20. JAVELIN Renal 101: only IO combo without OS benefit.
21. All modern first-line IO trials used sunitinib as comparator.
22. Chemotherapy response in ccRCC: 5.5–6%.
23. Collecting duct carcinoma: gemcitabine + carboplatin.
24. Spontaneous regression after nephrectomy: <1%.
25. Metastasectomy 5-yr survival after complete resection: 35–50%.
26. SURTIME median OS favoured deferred CN: 32.4 vs 15.0 mo.
27. CARMENA: ~45% MSKCC poor risk.
28. Tivozanib: after ≥2 prior systemic therapies (TIVO-3).
29. Belzutifan also approved for VHL-associated RCC.
30. Ipi + nivo dosing: nivo 3 mg/kg + ipi 1 mg/kg q3w ×4.
31. Active surveillance listed only for favourable risk.
32. Autoimmune events with ipilimumab predicted response.
33. Bevacizumab (anti-VEGF-A): first VEGF-pathway antagonist tested in ccRCC.
34. Sarcomatoid tumours: more PD-1+ TILs; CLEAR benefit held regardless of sarcomatoid histology.

★ Cross-chapter tables & night-before sheet

Reading time ≈ 12 min (slow pace) · Integrates Ch 136–144 · the frames most asked as long essays

A · Solid renal mass: choosing the treatment

	Active surveillance	Thermal ablation	Partial nephrectomy	Radical nephrectomy
Best for	Elderly, frail, life expectancy <5 y; solid <2 cm or cystic	cT1a ≤3 cm, comorbid; biopsy first	cT1a standard; imperative (solitary, bilateral, familial, CKD)	High complexity + no CKD/proteinuria + expected GFR >45 (all three)
Key numbers	Growth 0.1–0.3 cm/y; mets <2%; triggers >3 cm or >5 mm/y	Cryo ≤−40 °C target, 2 freeze-thaw; RFA/MWA >60–70 °C	Trifecta WIT ≤25 min; margin width immaterial	New GFR ≈ GFR × contralateral SRF × 1.25
Weakness	Needs imaging adherence	Higher local recurrence (re-ablete); no pathology	Urine leak, bleeding, pseudoaneurysm	CKD 65% vs 20% after PN
Chapters	137	143	137, 141, 142	137, 141, 142

B · Renal lesion at a glance

Lesion	Clue	Genetics / IHC	Default action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	Size-based: ≤2 cm AS; >4 cm excise
AML	Fat ≤−20 HU; fat-poor = trap	TSC2/TSC1; HMB-45, Melan-A	>6 cm / aneurysm >5 mm / bleed → SAE; TSC → everolimus
Oncocytoma	Stellate scar, segmental enhancement inversion	KIT+, CK7 single-cell; BHD	Biopsy cannot exclude chRCC → AS or PN
ccRCC	Hypervascular, heterogeneous	3p/VHL, HIF-2α; CA-IX	PN / RN / TA / AS
Papillary RCC	Hypoenhancing	+7, +17, −Y; MET	As RCC
Chromophobe	Moderate enhancement	Multiple losses; CK7 diffuse	As RCC; good prognosis
Lymphoma / metastasis	Infiltrative, hypovascular, adenopathy	—	Biopsy, don't operate
UTUC	Filling defect, ~46 HU, positive cytology	MSI → Lynch	Risk-stratify → KSM or RNU

C · UTUC: kidney-sparing vs RNU

	Kidney-sparing (URS / PCNL / segmental)	RNU + bladder cuff
Who	Low risk (ALL: unifocal, <2 cm, LG cytology & biopsy, no invasion on CTU); imperative (solitary, CKD, bilateral)	High risk (ANY: multifocal, >2 cm, HG, hydronephrosis, prior cystectomy, variant)
Extras	URS every visit; UGN-101 for LG; ~3× reinterventions	Within 12 wk; LND template; single-dose intravesical chemo post-op
Systemic	—	Neoadjuvant platinum if CrCl will fall <50; adjuvant (POUT) for ≥pT2/N+

D · Open vs minimally invasive kidney surgery

Choose open (Ch 141)	Choose lap/robotic (Ch 142)
Locally advanced RCC (esp. IVC thrombus needing bypass/CPB); cytoreductive setting; anatomically complex tumours; significant prior abdominal surgery. Trauma: midline, early vascular control	Standard for RN of organ-confined tumours and most PN; less pain, blood loss, stay; retroperitoneal = shorter operating time

E · Metastatic RCC in four lines

IMDC (6): KPS <80, Hb <LLN, Ca >ULN, neutrophils >ULN, platelets >ULN, <1 y to systemic Rx → 0 fav / 1-2 int / ≥3 poor

First line: IO + TKI (pembro+axi, nivo+cabo, pembro+lenva) any risk; ipi+nivo for intermediate/poor

CN: upfront only if fit, low burden, favourable; else systemic first → deferred CN in responders (CARMENA, SURTIME)

Later lines: switch mechanism; belzutifan after IO + TKI

F · Night-before sheet: 40 must-know facts

- Simple cyst ≤20 HU; enhancement >15-20 HU = suspicious.
- Bosniak III ~50%, IV ~90% malignant; IIF surveillance up to 5 y.
- AML fat ≤-20 HU; treat >6 cm, aneurysm >5 mm, pregnancy plans.
- TSC AML >3 cm → everolimus (EXIST-2).
- ~20% of cT1 masses benign; ~40% of ≤2 cm.
- ccRCC: 3p/VHL; papillary: +7/+17; chromophobe: multiple losses.
- VHL 3p25 · MET 7q31 (only oncogene) · FH 1q42 · FLCN 17p11.2.
- 3-cm rule for VHL/HPRC/BHD; HLRCC → excise any size.
- Belzutifan = HIF-2α inhibitor (VHL disease; post-IO/TKI).
- Commonest paraneoplastic: ↑ESR; only medically treatable: hypercalcaemia.
- Stauffer = non-metastatic hepatic dysfunction (IL-6).
- ISUP grade 4 = sarcomatoid / rhabdoid / giant cells.
- T1a ≤4 · T1b 4-7 · T2a 7-10 · T2b >10 cm.
- T3a renal vein / sinus fat / pelvicalyceal; T3b IVC below, T3c above diaphragm.
- Contiguous adrenal = T4; non-contiguous = M1.
- Thrombus levels: I ≤2 cm above ostium; II below hepatics; III intrahepatic; IV supradiaphragmatic.
- Level IV → CPB ± DHCA (16-18 °C); Pringle ≤60 min.
- RN preferred only when high complexity + no CKD + GFR likely >45.
- CKD after RN 65% vs PN 20%; CKD-surgical is stable.
- EORTC 30904: PN vs RN; RN better OS (flawed).
- EORTC 30881: LND no survival benefit.
- KEYNOTE-564: adjuvant pembrolizumab, DFS HR 0.63.
- S-TRAC: only positive adjuvant TKI (no OS gain).
- CARMENA: sunitinib alone non-inferior to CN + sunitinib.
- SURTIME: deferred CN safe.
- CheckMate 214: ipi+nivo beats sunitinib in intermediate/poor risk.
- TA for cT1a ≤3 cm; cryo 2 freeze-thaw cycles; RFA needs grounding pads.
- Post-ablation central/nodular enhancement = recurrence.
- Trifecta: negative margin, no complications, WIT ≤25 min.
- Urine leak: drain creatinine > serum; most settle in 4-6 wk.
- Delayed haematuria after PN → pseudoaneurysm → angioembolisation.
- UTUC high risk: any one of multifocal, >2 cm, HG, hydronephrosis, variant, prior cystectomy.
- UTUC bladder recurrence 22-47% → cystoscopy for all.
- Lynch: MSH2 highest UTUC risk; MSI flags Lynch.
- Aristolochic acid: A:T→T:A transversion; Balkan nephropathy.
- Retained intramural ureteric stump: 30-75% recurrence.
- ODMIT-C: single MMC dose post-RNU ↓ bladder recurrence.
- POUT: adjuvant platinum, DFS HR 0.45.
- UGN-101 (Jelmyto): MMC gel for LG UTUC; stricture ~44%.
- Endoscopic cut-off (AUA/SUO): <1.5 cm; Ho:YAG penetration <0.5 mm.